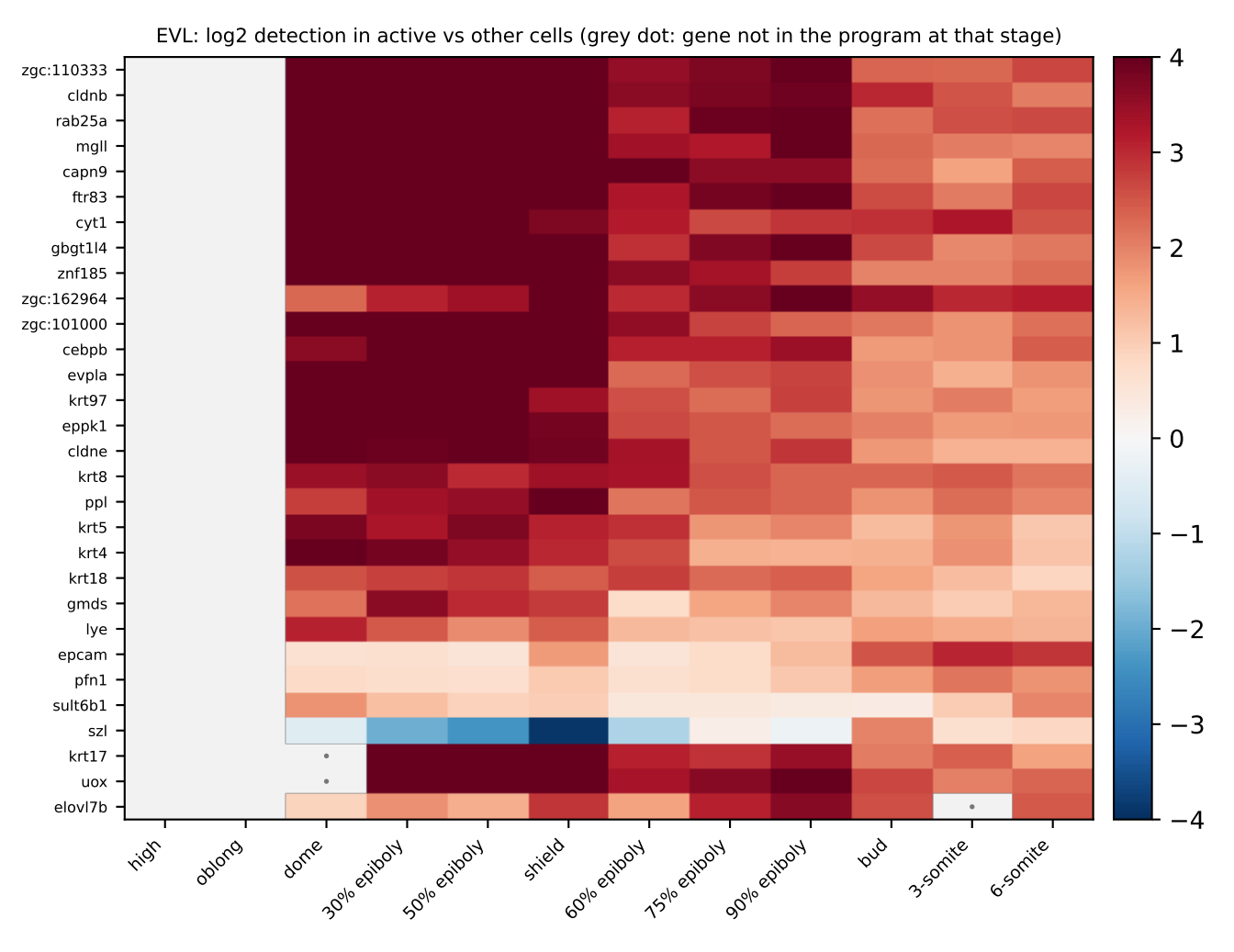
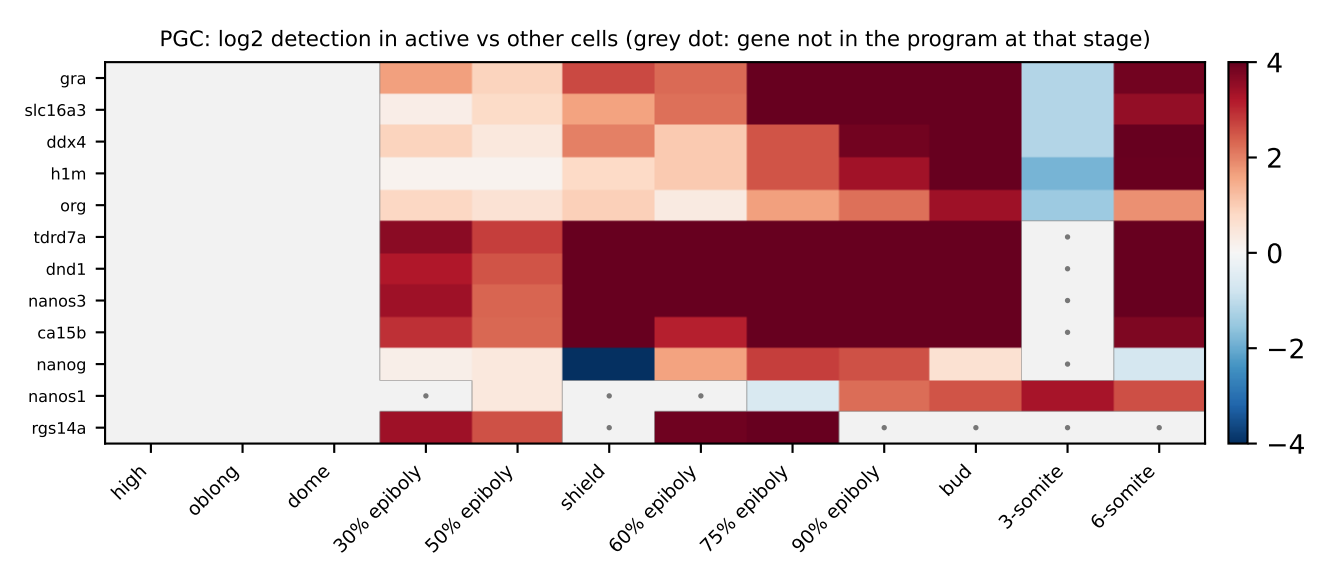
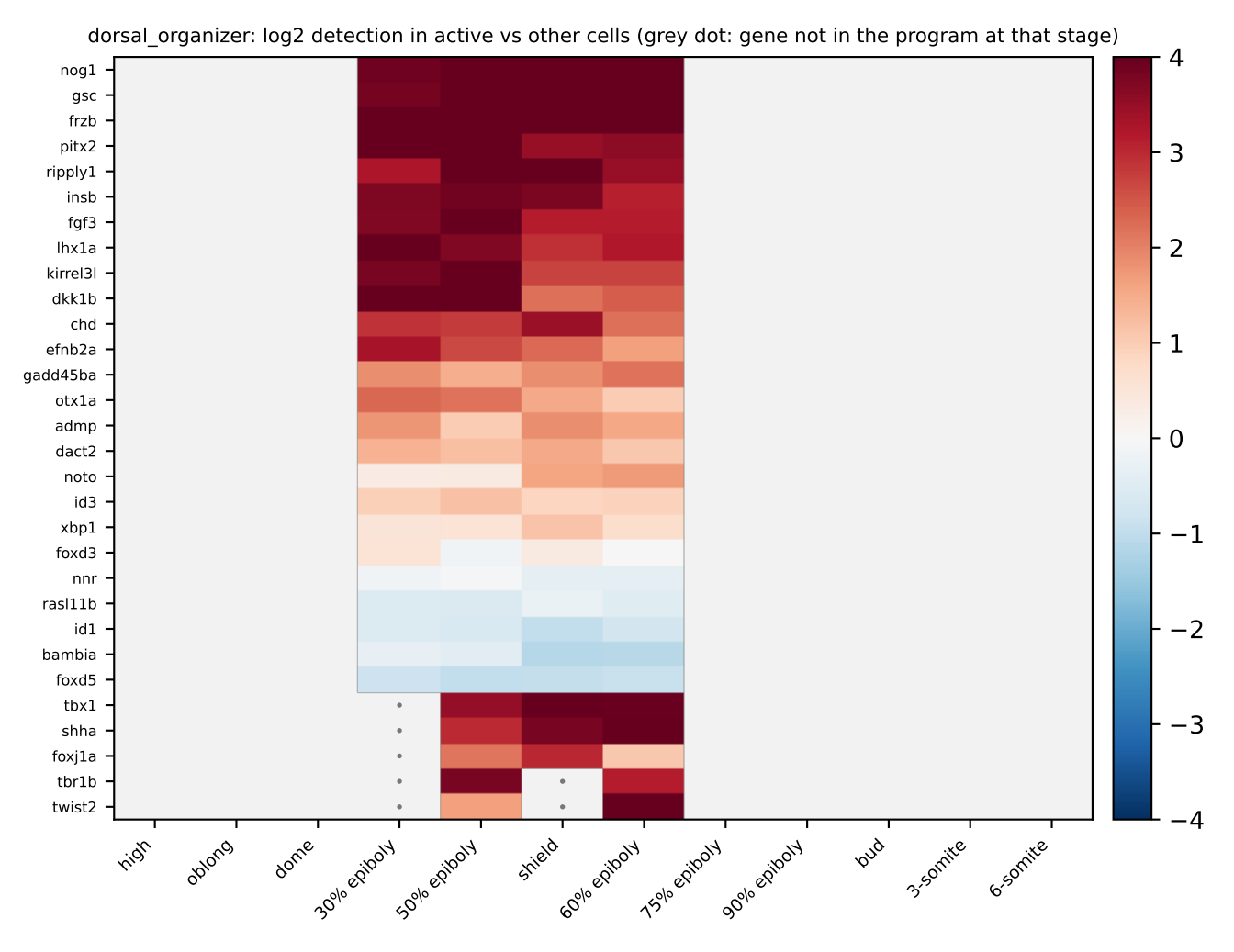
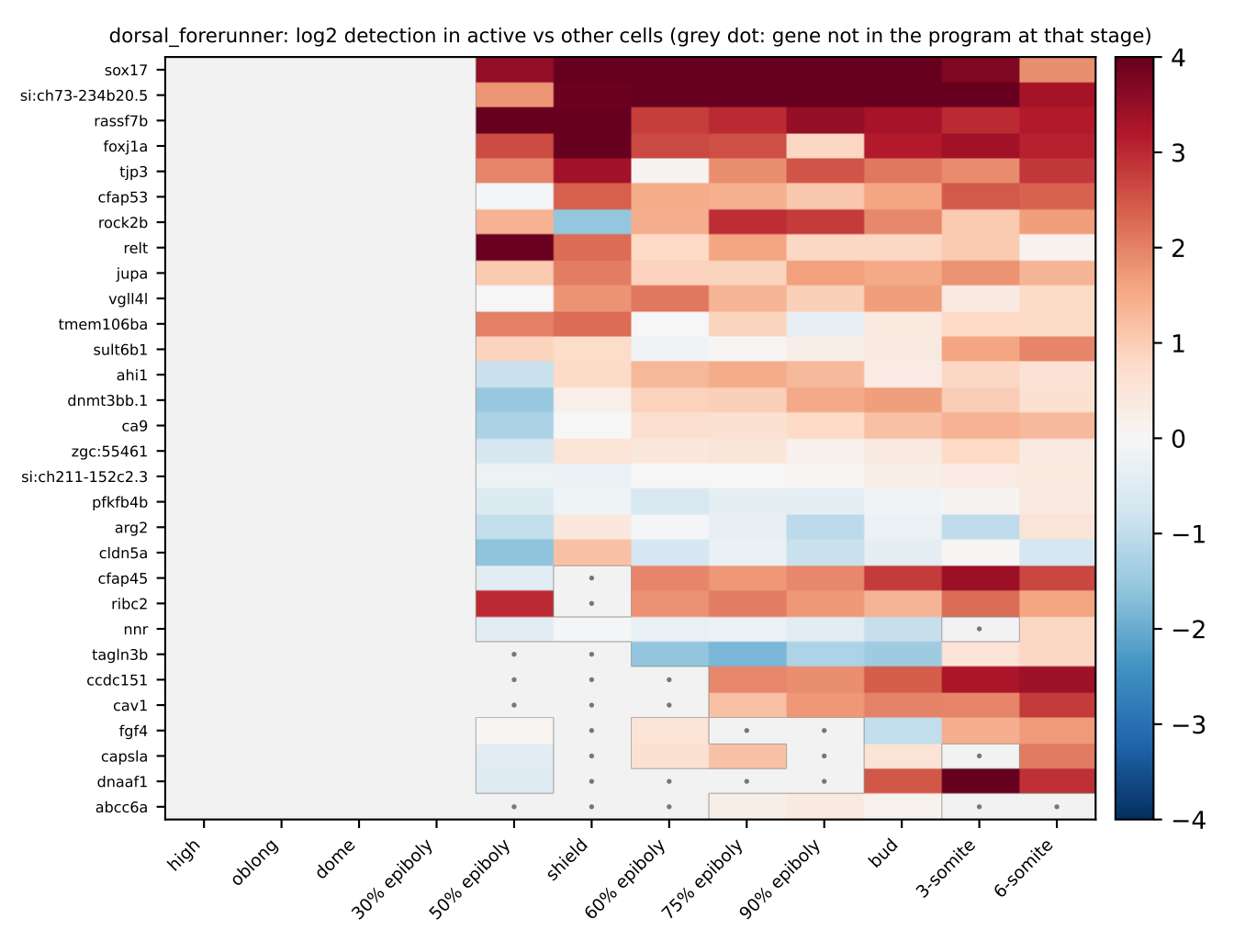
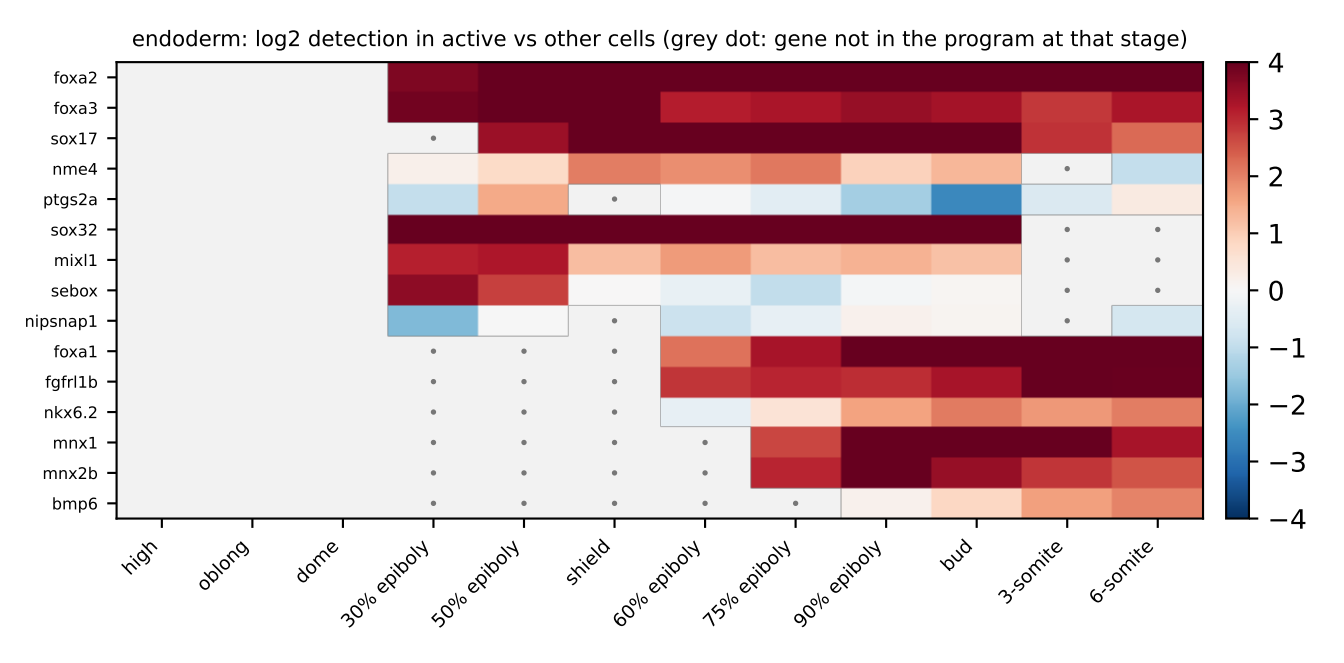
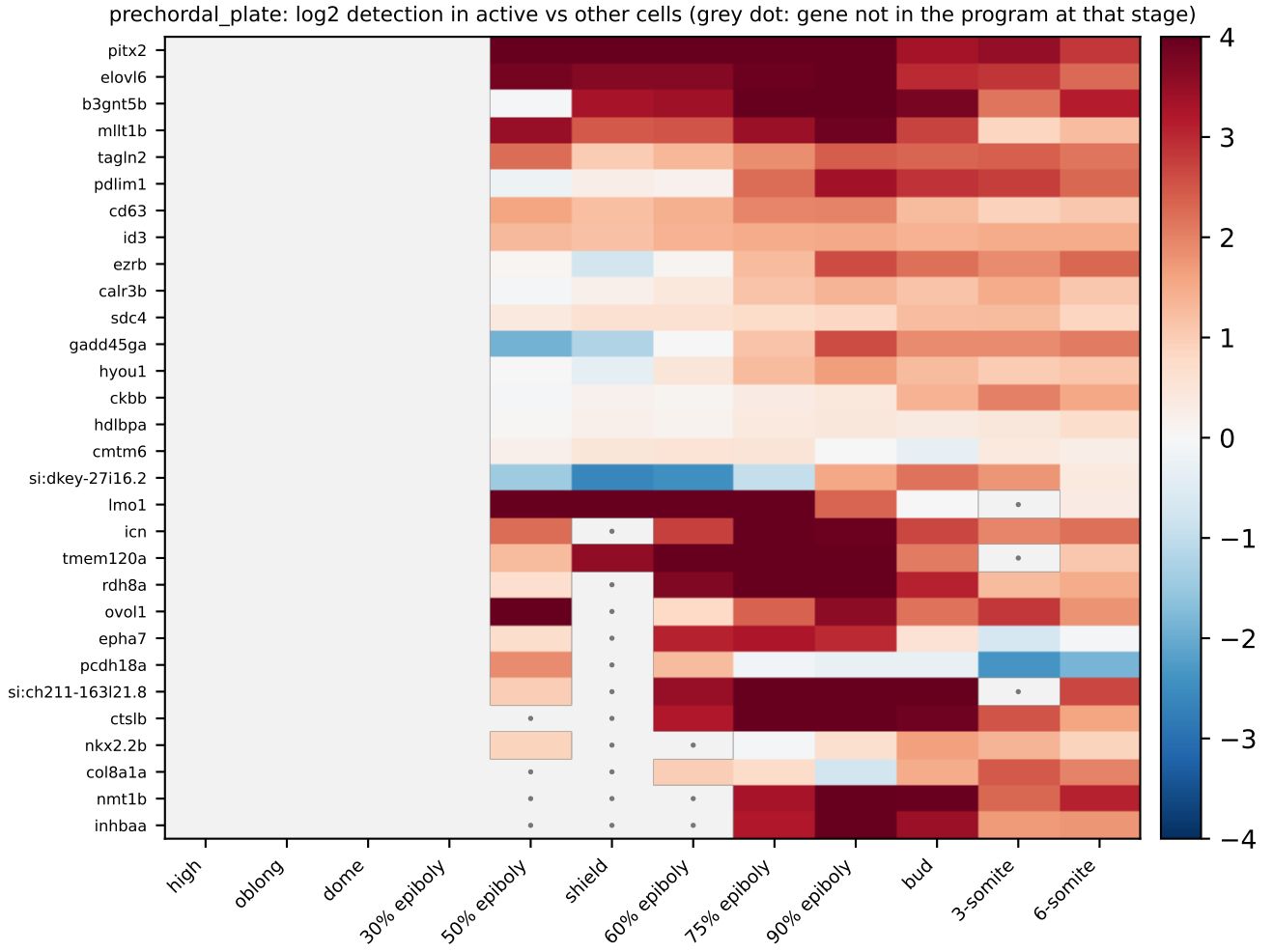

EVL: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
zgc:110333
cldnb
rab25a
mgll
capn9
ftr83
cyt1
gbgt1l4
znf185
zgc:162964
zgc:101000
cebpb
evpla
krt97
eppk1
cldne
krt8
ppl
krt5
krt4
krt18
gmds
lye
epcam
pfn1
sult6b1
szl
krt17
uox
elovl7b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

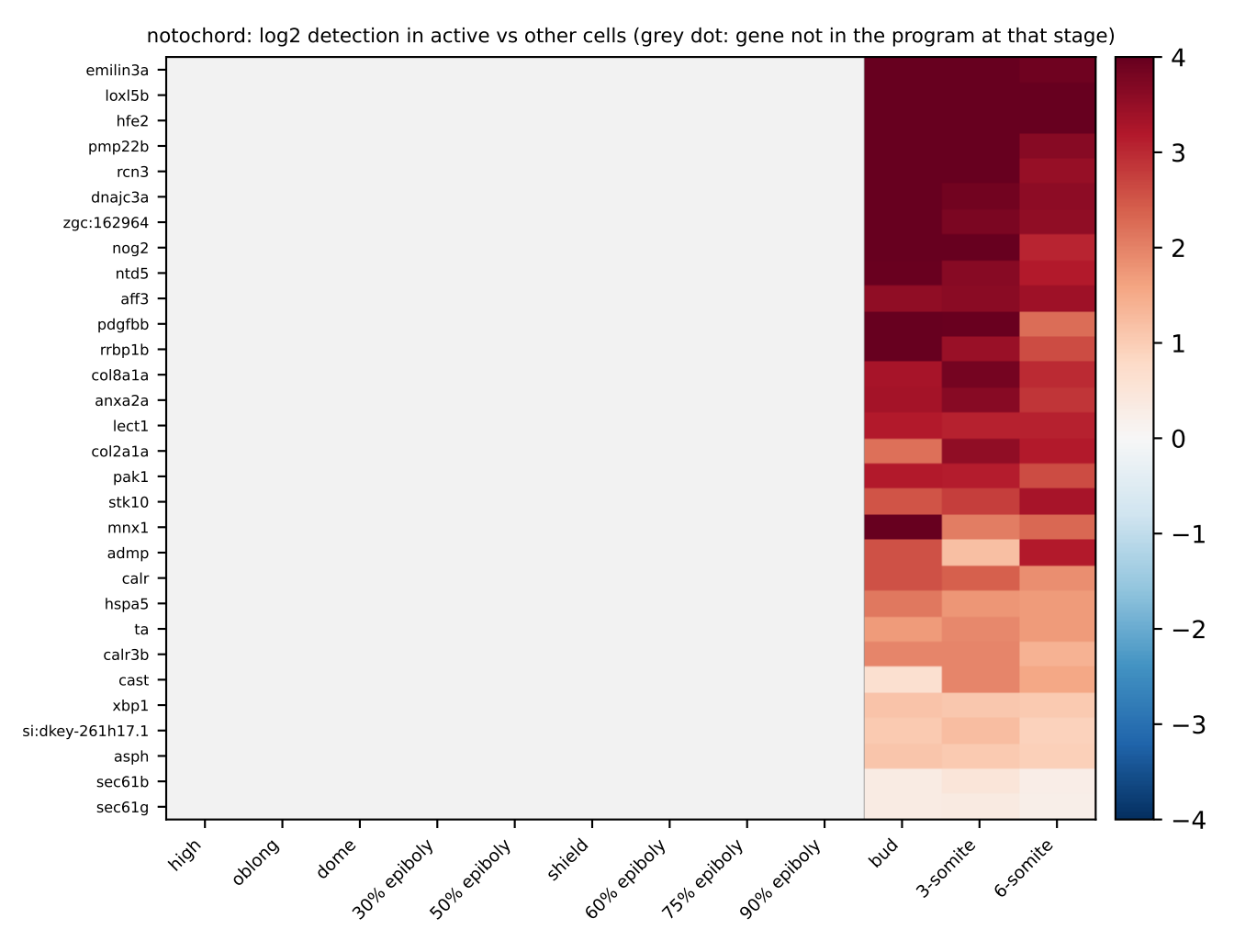

PGC: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gra
slc16a3
ddx4
h1m
org
tdrd7a
dnd1
nanos3
ca15b
nanog
nanos1
rgs14a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

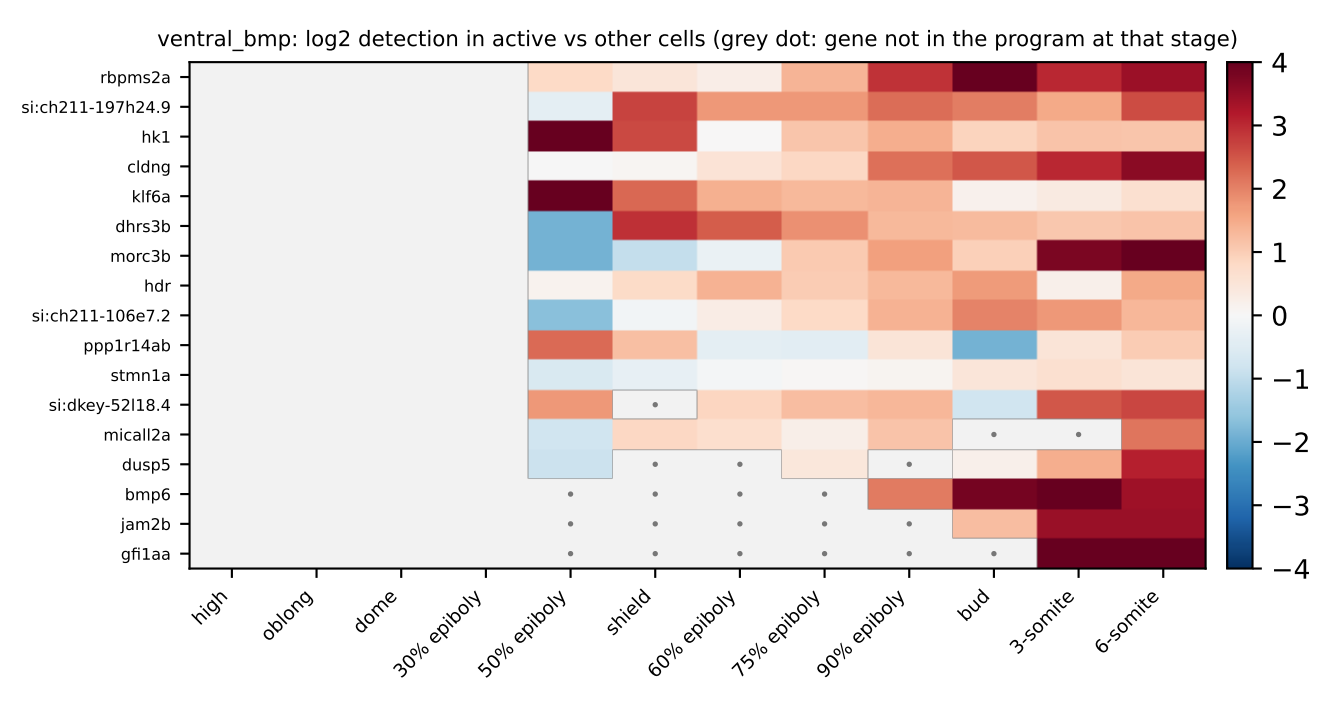

dorsal_organizer: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
nog1
gsc
frzb
pitx2
ripply1
insb
fgf3
lhx1a
kirrel3l
dkk1b
chd
efnb2a
gadd45ba
otx1a
admp
dact2
noto
id3
xbp1
foxd3
nnr
rasl11b
id1
bambia
foxd5
tbx1
shha
foxj1a
tbr1b
twist2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

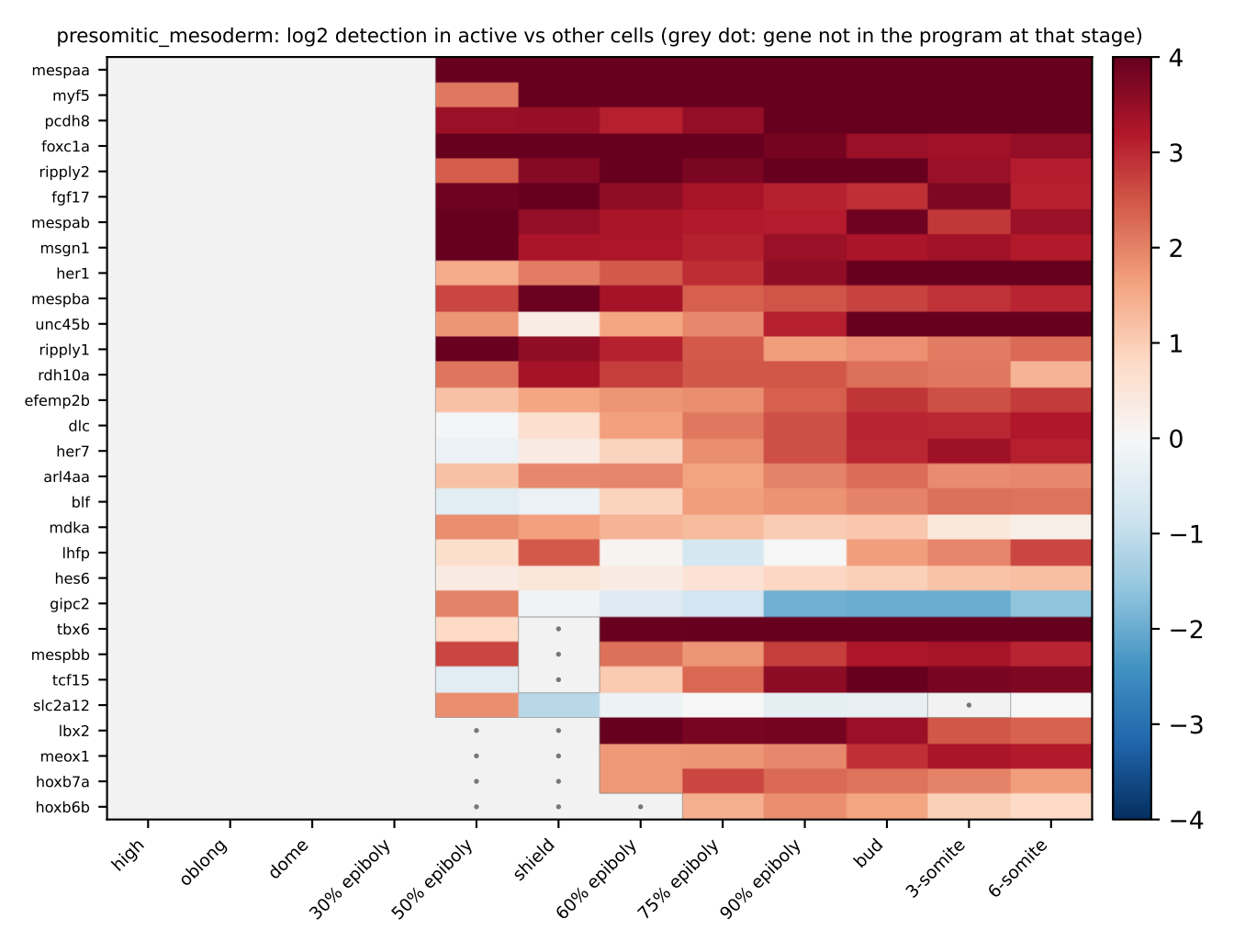

dorsal_forerunner: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
sox17
si:ch73-234b20.5
rassf7b
foxj1a
tjp3
cfap53
rock2b
relt
jupa
vgll4l
tmem106ba
sult6b1
ahi1
dnmt3bb.1
ca9
zgc:55461
si:ch211-152c2.3
pfkfb4b
arg2
cldn5a
cfap45
ribc2
nnr
tagln3b
ccdc151
cav1
fgf4
capsla
dnaaf1
abcc6a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

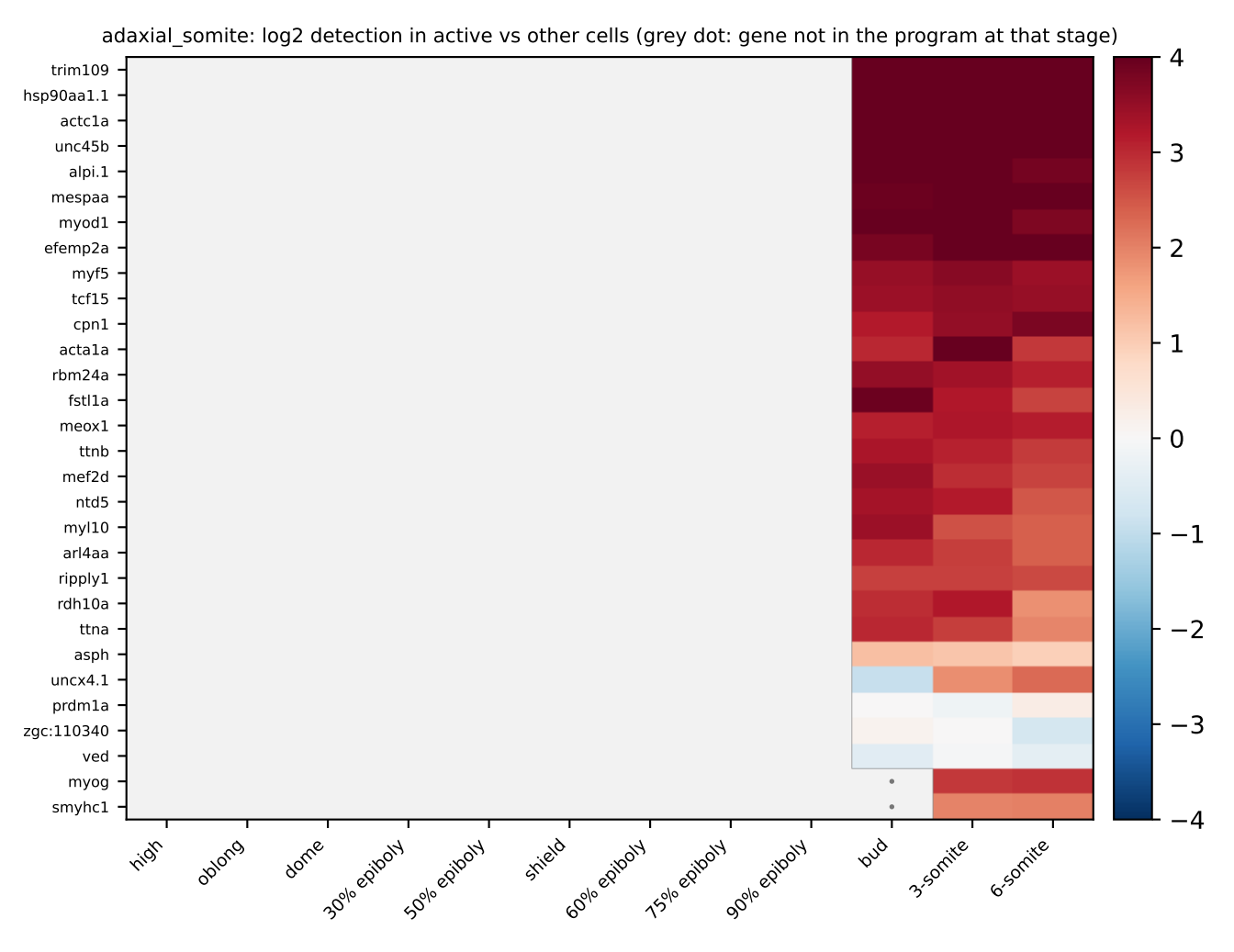

endoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
foxa2
foxa3
sox17
nme4
ptgs2a
sox32
mixl1
sebox
nipsnap1
foxa1
fgfrl1b
nkx6.2
mnx1
mnx2b
bmp6
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

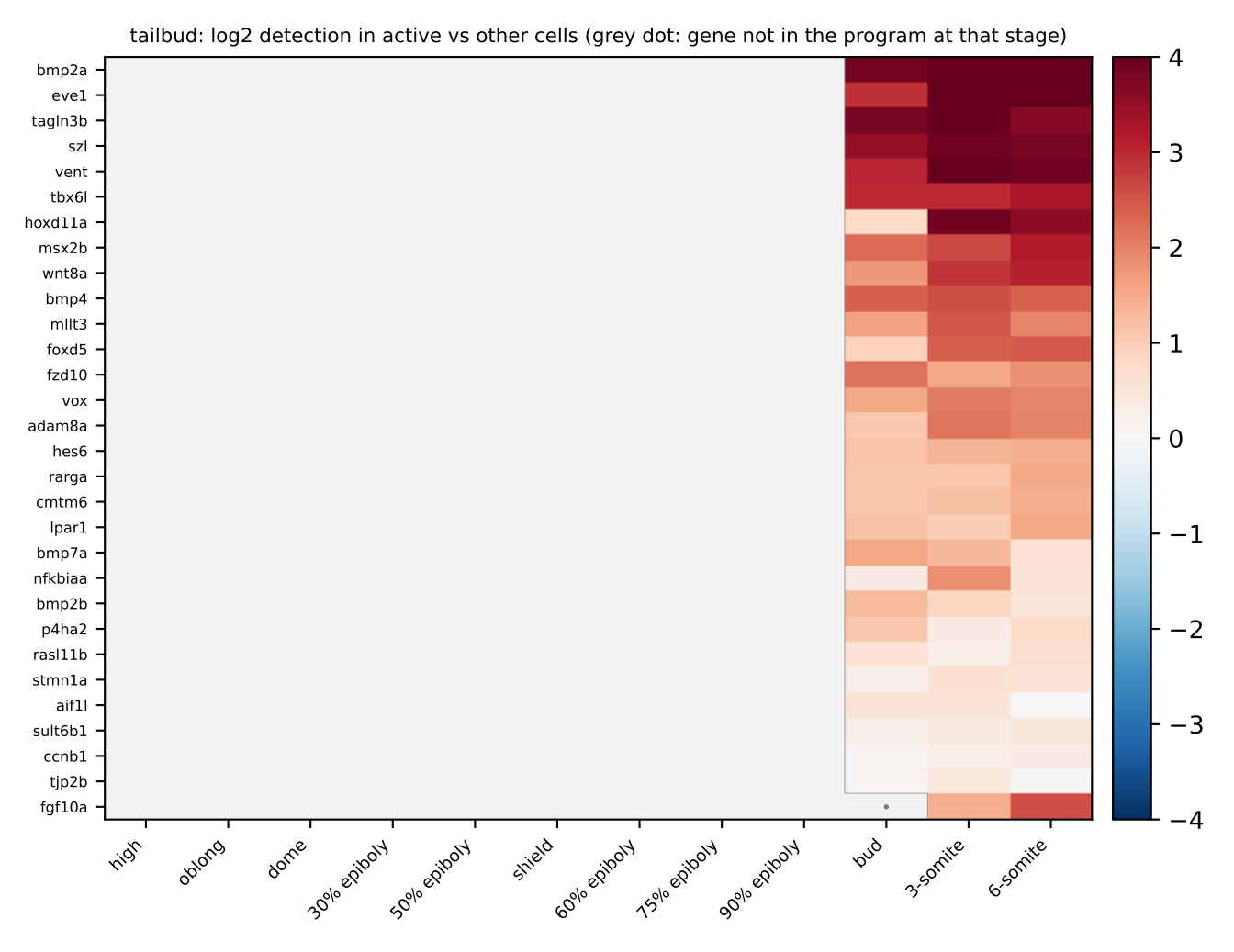

prechordal_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
pitx2
elovl6
b3gnt5b
mllt1b
tagln2
pdlim1
cd63
id3
ezrb
calr3b
sdc4
gadd45ga
hyou1
ckbb
hdlbpa
cmtm6
si:dkey-27i16.2
lmo1
icn
tmem120a
rdh8a
ovol1
epha7
pcdh18a
si:ch211-163l21.8
ctslb
nkx2.2b
col8a1a
nmt1b
inhbaa
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

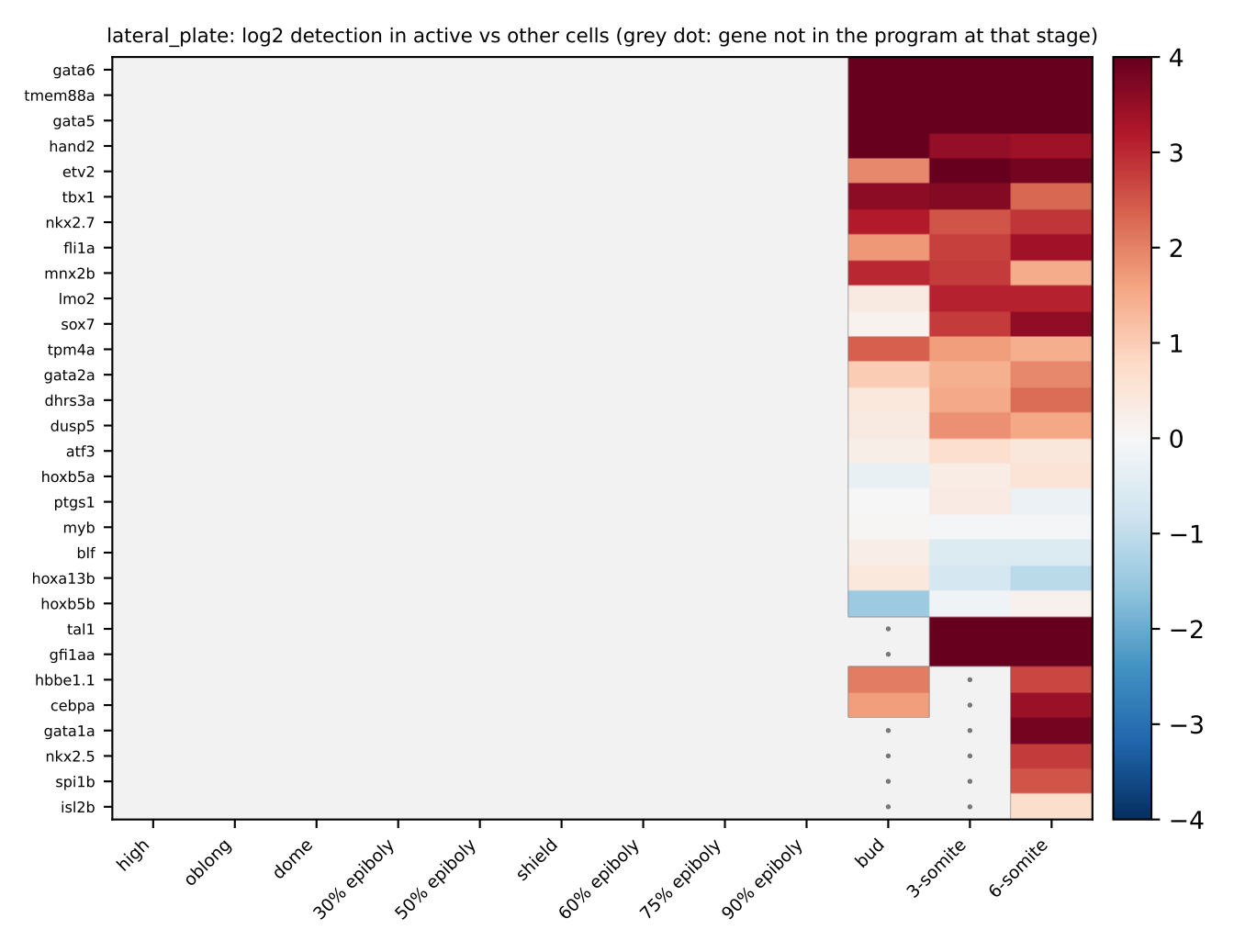

notochord: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
emilin3a
loxl5b
hfe2
pmp22b
rcn3
dnajc3a
zgc:162964
nog2
ntd5
aff3
pdgfbb
rrbp1b
col8a1a
anxa2a
lect1
col2a1a
pak1
stk10
mnx1
admp
calr
hspa5
ta
calr3b
cast
xbp1
si:dkey-261h17.1
asph
sec61b
sec61g
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

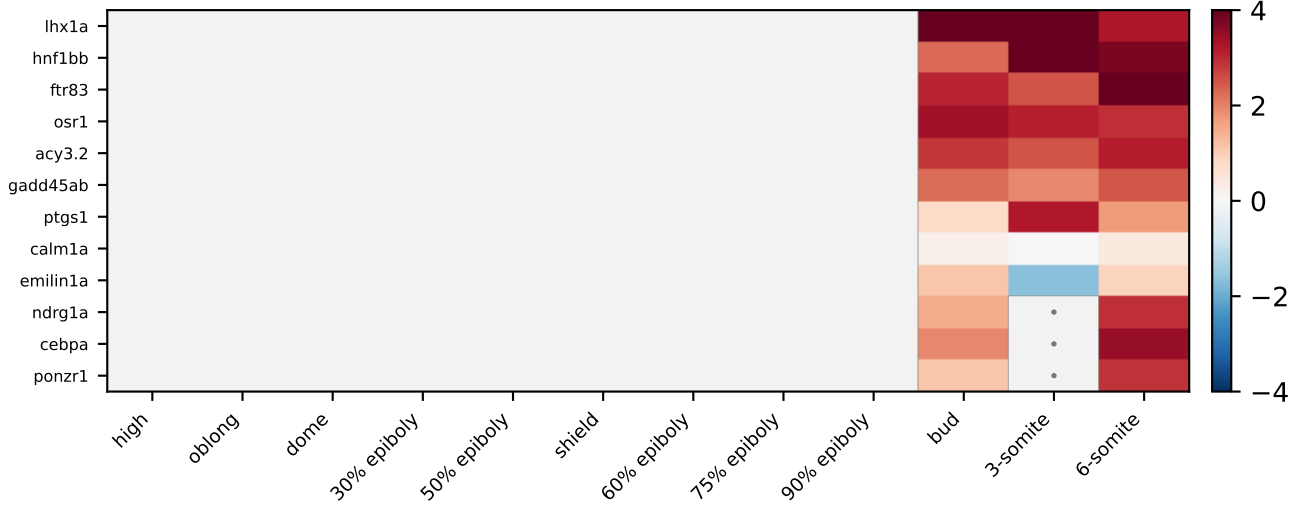

ventral_bmp: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
rbpms2a
si:ch211-197h24.9
hk1
cldng
klf6a
dhrs3b
morc3b
hdr
si:ch211-106e7.2
ppp1r14ab
stmn1a
si:dkey-52l18.4
micall2a
dusp5
bmp6
jam2b
gfi1aa
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

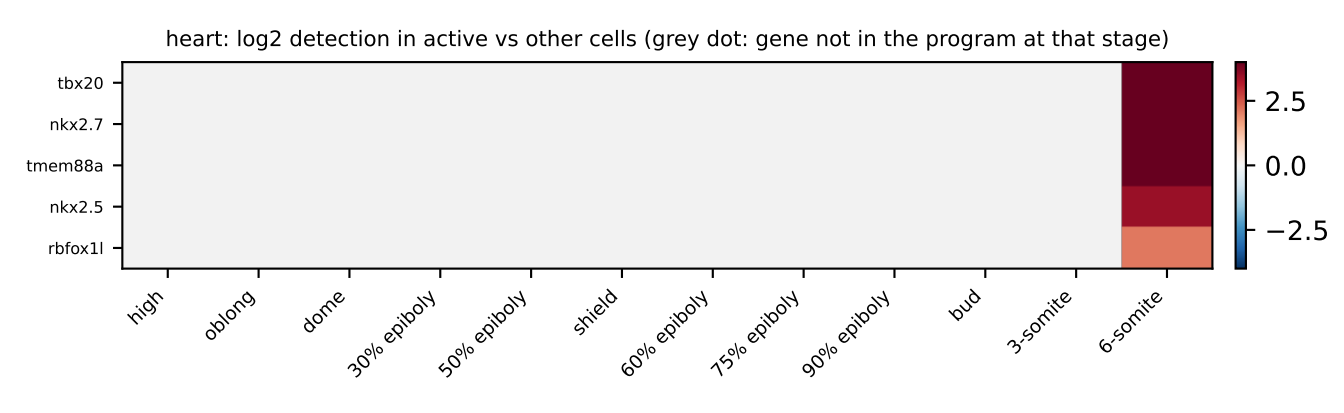

presomitic_mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
mespaa
myf5
pcdh8
foxc1a
ripply2
fgf17
mespab
msgn1
her1
mespba
unc45b
ripply1
rdh10a
efemp2b
dlc
her7
arl4aa
blf
mdka
lhfp
hes6
gipc2
tbx6
mespbb
tcf15
slc2a12
lbx2
meox1
hoxb7a
hoxb6b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

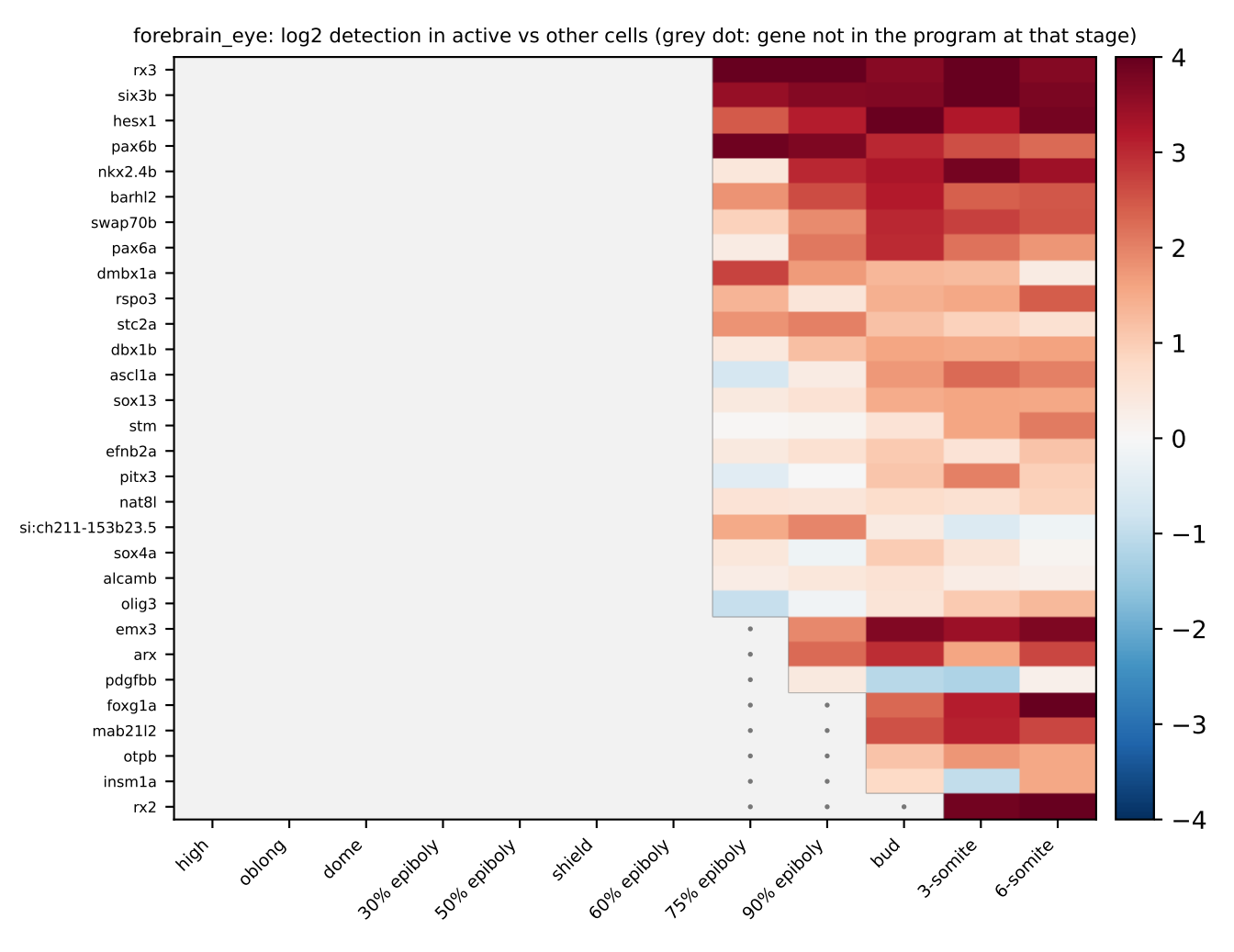

adaxial_somite: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
trim109
hsp90aa1.1
actc1a
unc45b
alpi.1
mespaa
myod1
efemp2a
myf5
tcf15
cpn1
acta1a
rbm24a
fstl1a
meox1
ttnb
mef2d
ntd5
myl10
arl4aa
ripply1
rdh10a
ttna
asph
uncx4.1
prdm1a
zgc:110340
ved
myog
smyhc1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

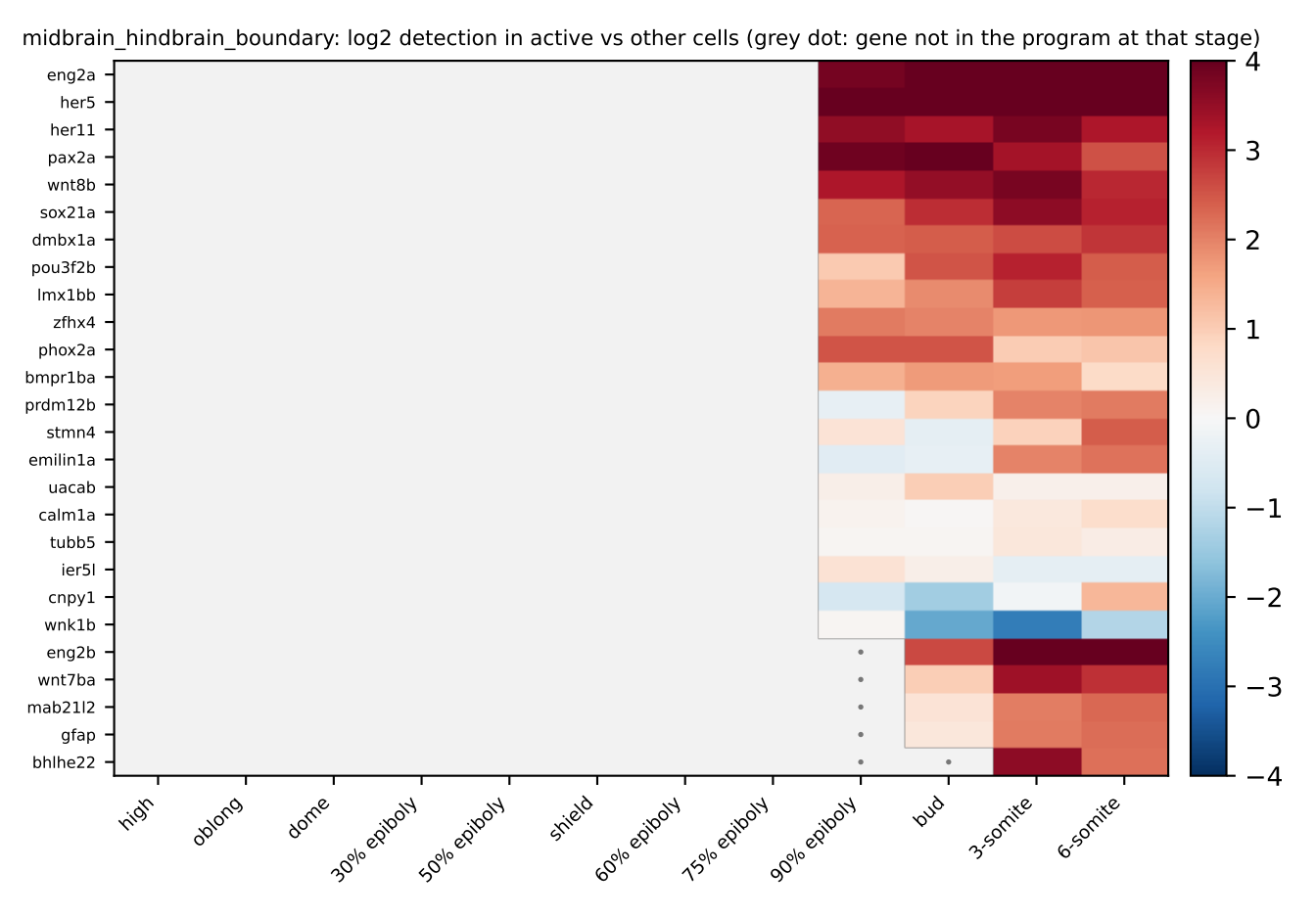

tailbud: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
bmp2a
eve1
tagln3b
szl
vent
tbx6l
hoxd11a
msx2b
wnt8a
bmp4
mllt3
foxd5
fzd10
vox
adam8a
hes6
rarga
cmtm6
lpar1
bmp7a
nfkbiaa
bmp2b
p4ha2
rasl11b
stmn1a
aif1l
sult6b1
ccnb1
tjp2b
fgf10a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

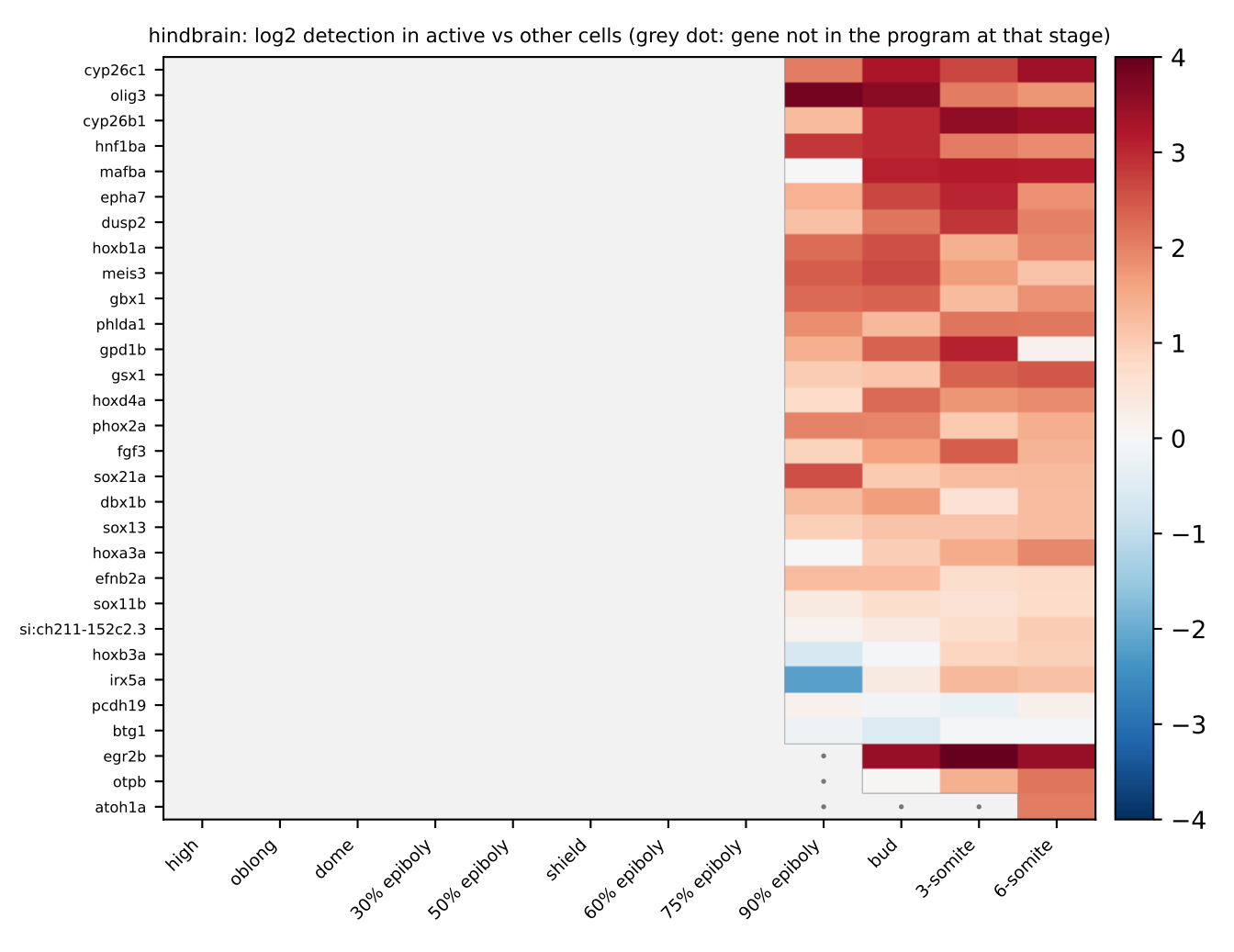

lateral_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gata6
tmem88a
gata5
hand2
etv2
tbx1
nkx2.7
fli1a
mnx2b
lmo2
sox7
tpm4a
gata2a
dhrs3a
dusp5
atf3
hoxb5a
ptgs1
myb
blf
hoxa13b
hoxb5b
tal1
gfi1aa
hbbe1.1
cebpa
gata1a
nkx2.5
spi1b
isl2b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

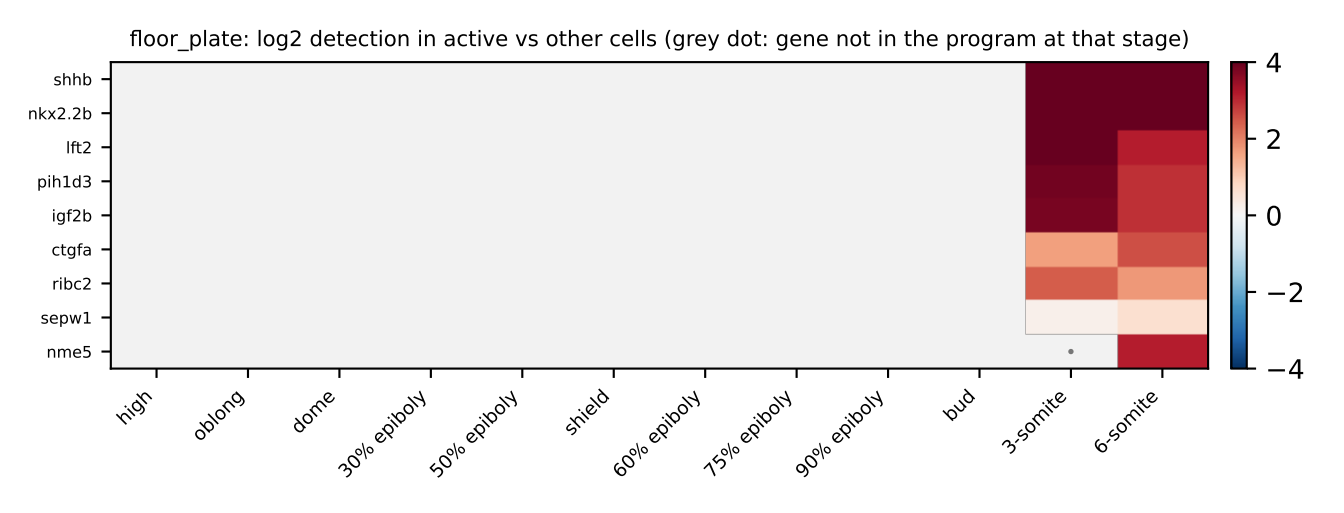

intermediate_mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
lhx1a
hnf1bb
ftr83
osr1
acy3.2
gadd45ab
ptgs1
calm1a
emilin1a
ndrg1a
cebpa
ponzr1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

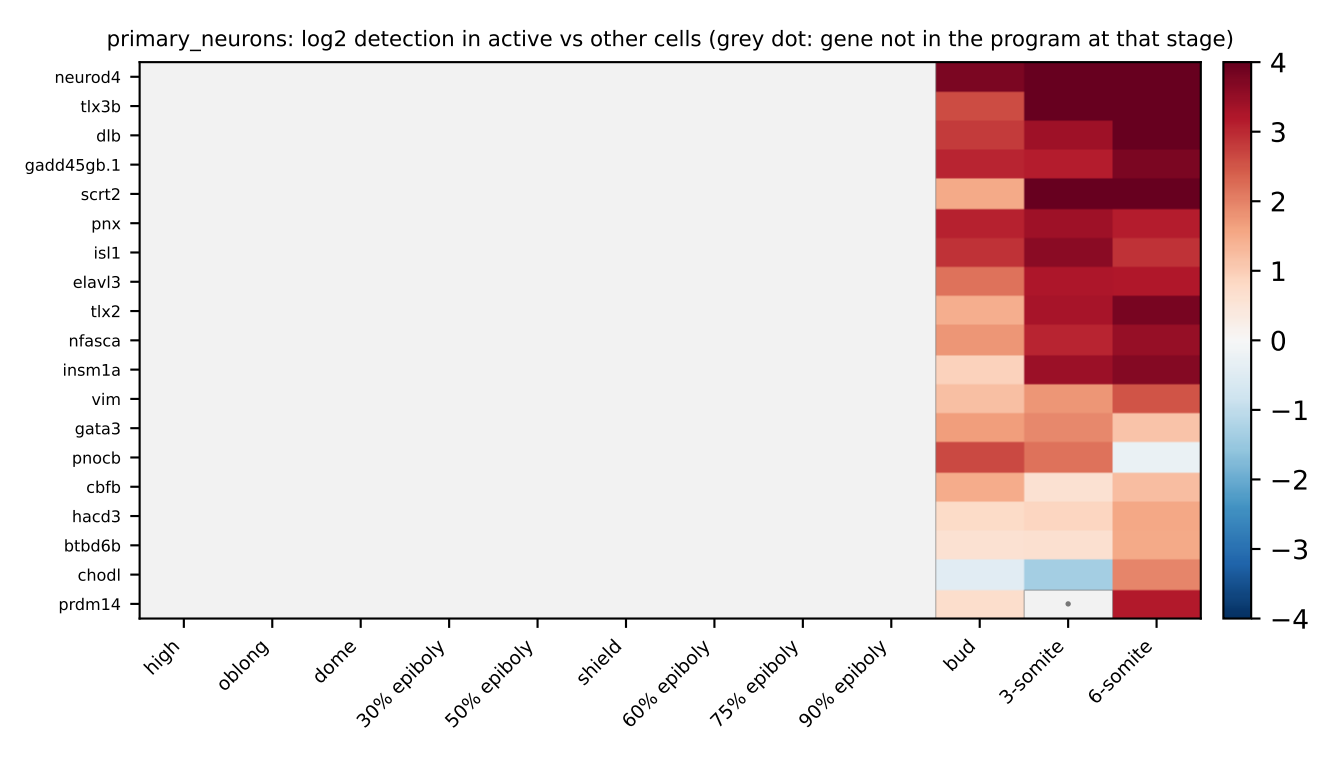

heart: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
tbx20
nkx2.7
tmem88a
nkx2.5
rbfox1l
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

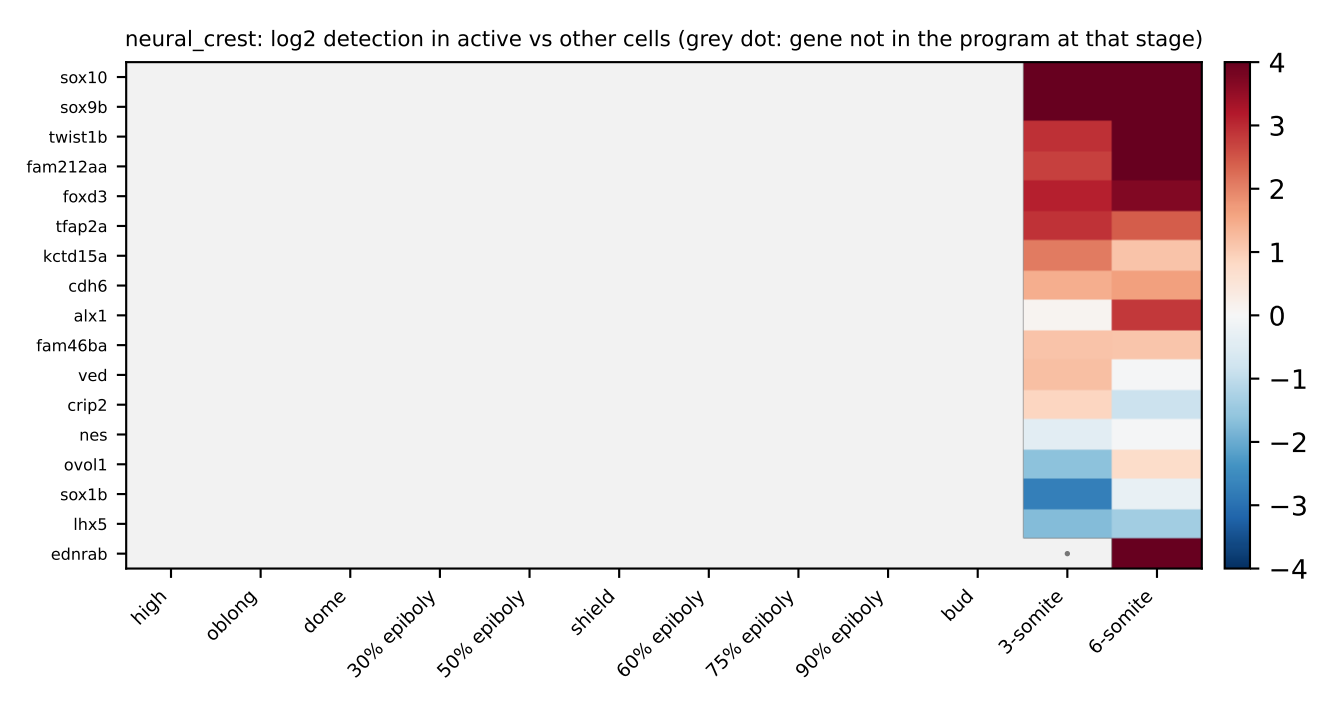

forebrain_eye: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
rx3
six3b
hesx1
pax6b
nkx2.4b
barhl2
swap70b
pax6a
dmbx1a
rspo3
stc2a
dbx1b
ascl1a
sox13
stm
efnb2a
pitx3
nat8l
si:ch211-153b23.5
sox4a
alcamb
olig3
emx3
arx
pdgfbb
foxg1a
mab21l2
otpb
insm1a
rx2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

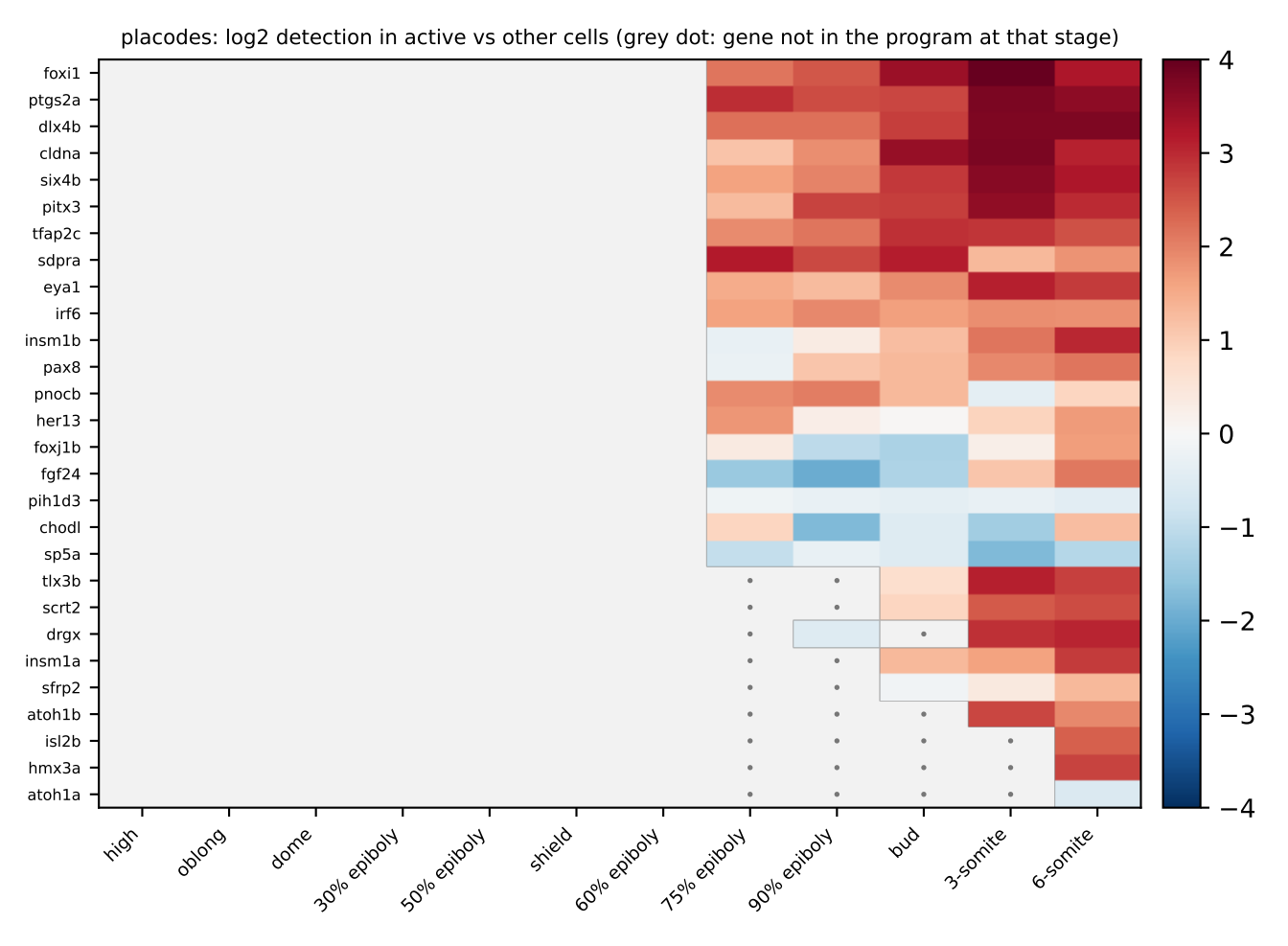

midbrain_hindbrain_boundary: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
eng2a
her5
her11
pax2a
wnt8b
sox21a
dmbx1a
pou3f2b
lmx1bb
zfhx4
phox2a
bmpr1ba
prdm12b
stmn4
emilin1a
uacab
calm1a
tubb5
ier5l
cnpy1
wnk1b
eng2b
wnt7ba
mab21l2
gfap
bhlhe22
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

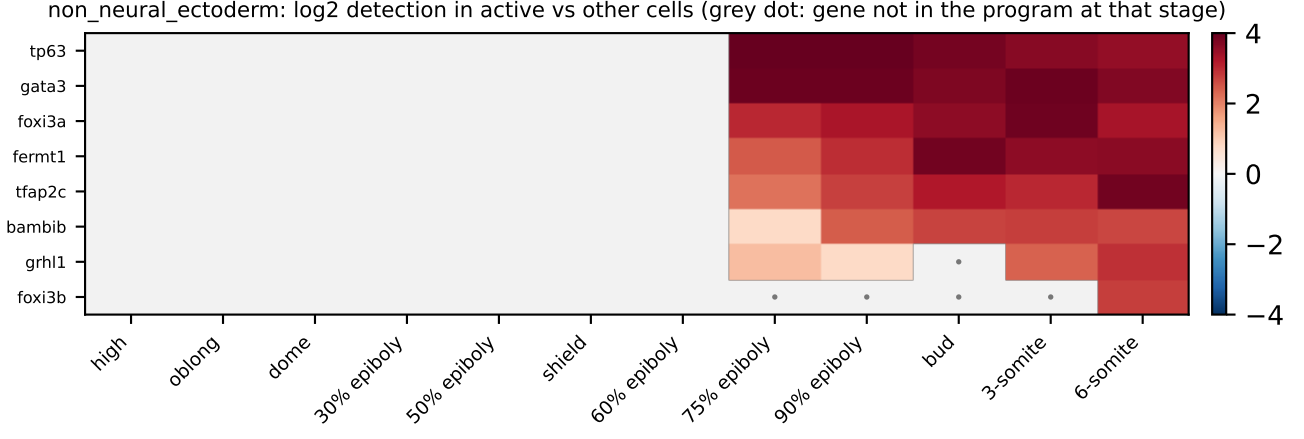

hindbrain: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
cyp26c1
olig3
cyp26b1
hnf1ba
mafba
epha7
dusp2
hoxb1a
meis3
gbx1
phlda1
gpd1b
gsx1
hoxd4a
phox2a
fgf3
sox21a
dbx1b
sox13
hoxa3a
efnb2a
sox11b
si:ch211-152c2.3
hoxb3a
irx5a
pcdh19
btg1
egr2b
otpb
atoh1a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

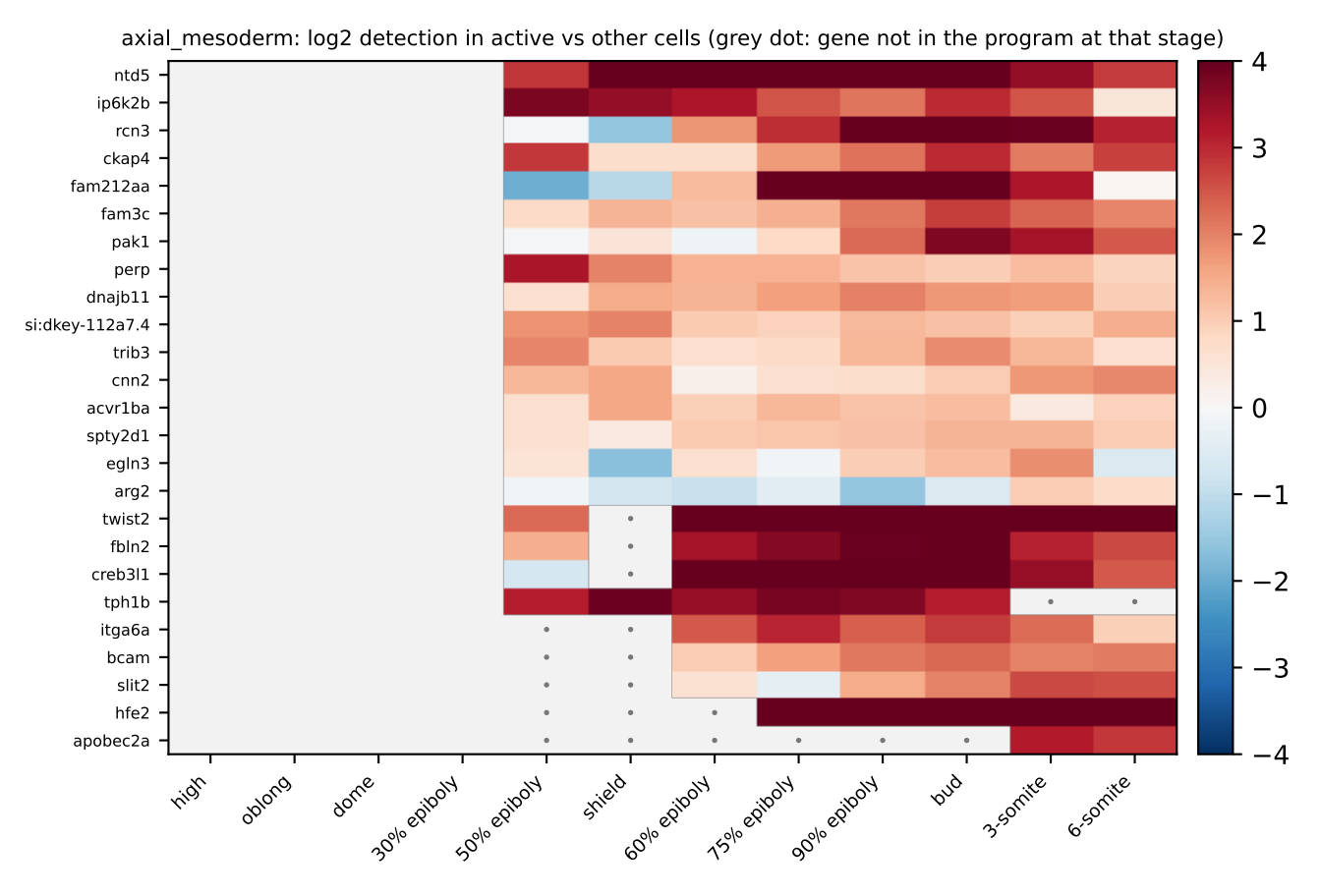

floor_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
shhb
nkx2.2b
lft2
pih1d3
igf2b
ctgfa
ribc2
sepw1
nme5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

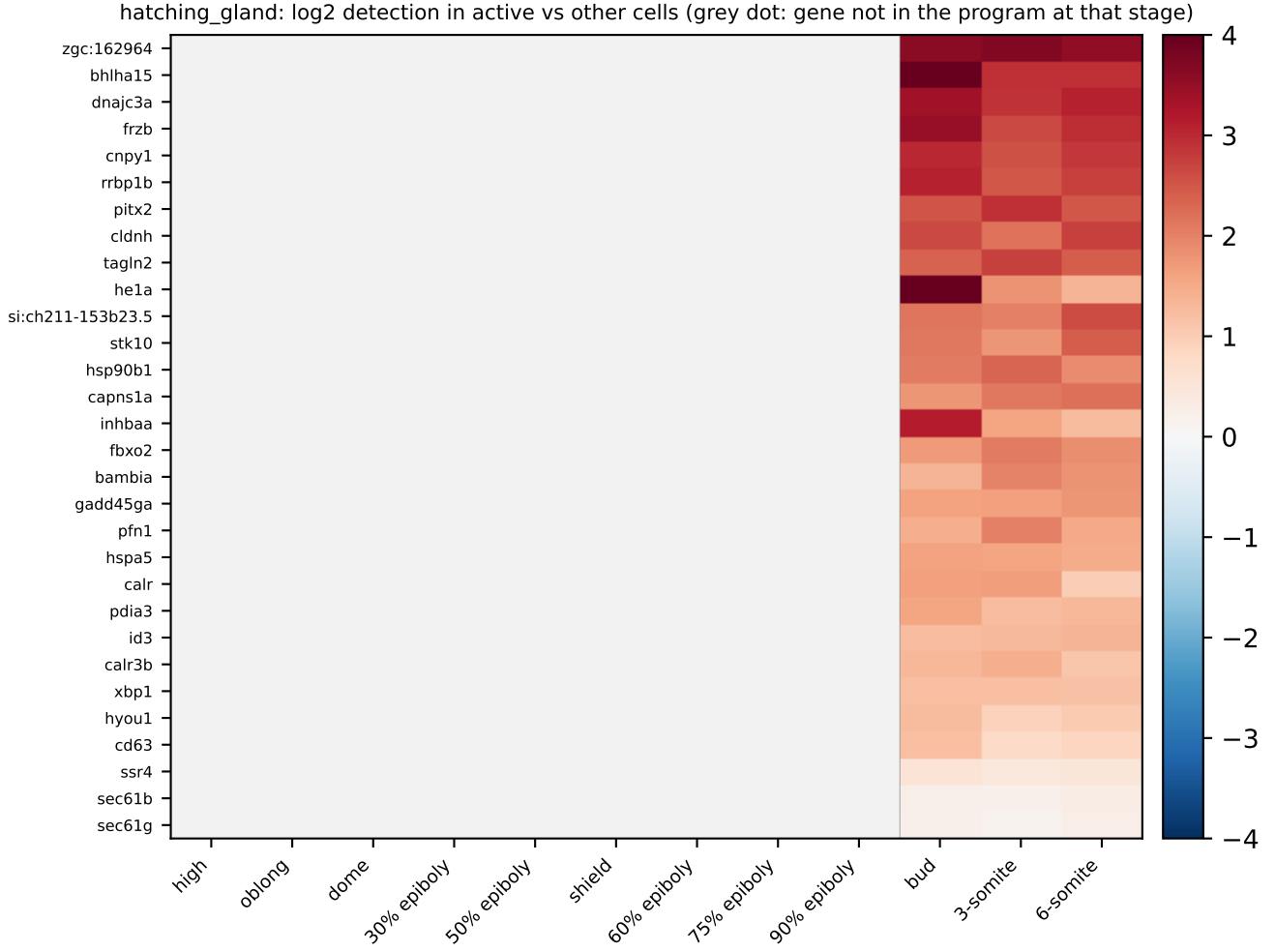

primary_neurons: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
neurod4
tlx3b
dlb
gadd45gb.1
scrt2
pnx
isl1
elavl3
tlx2
nfasca
insm1a
vim
gata3
pnocb
cbfb
hacd3
btbd6b
chodl
prdm14
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

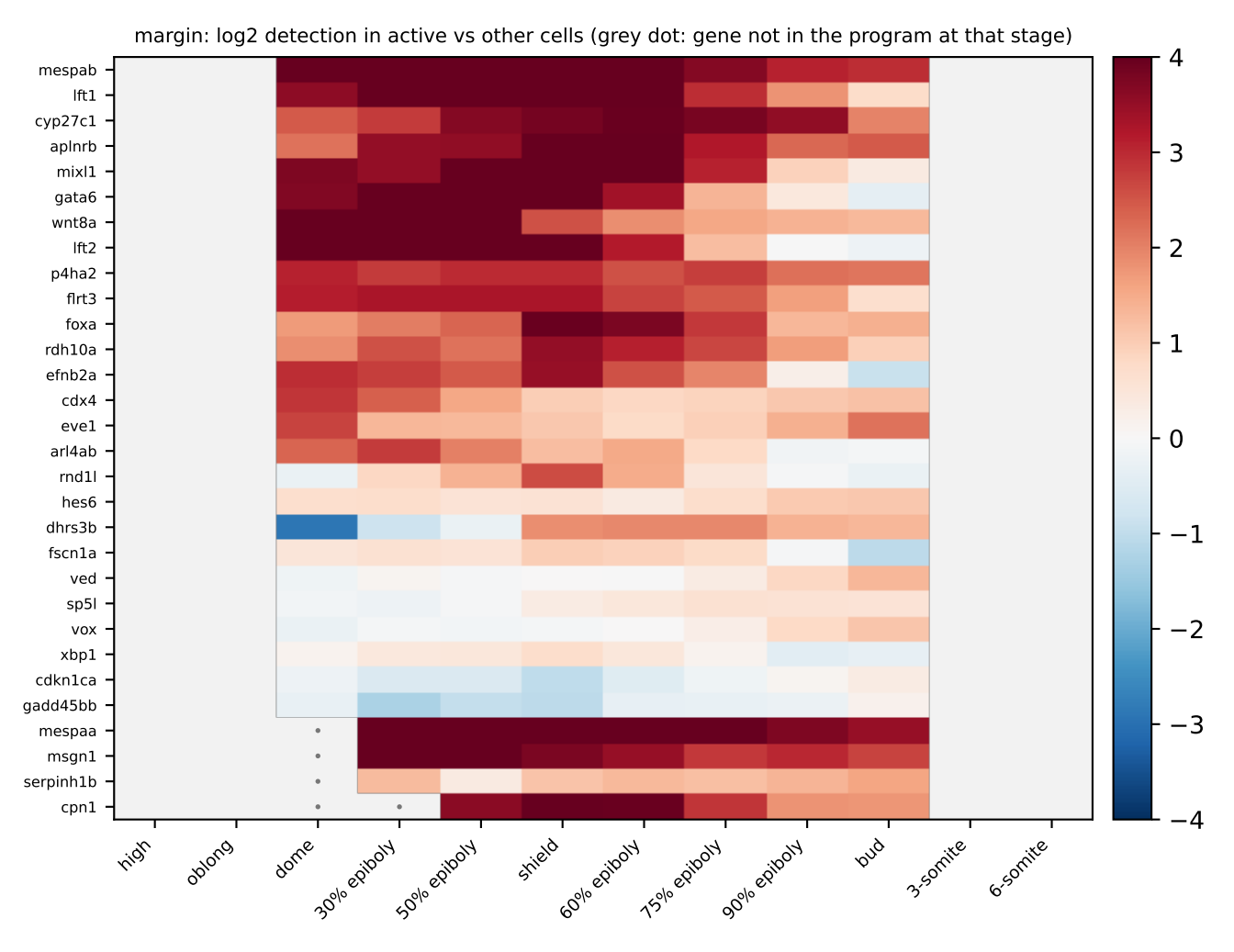

neural_crest: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
sox10
sox9b
twist1b
fam212aa
foxd3
tfap2a
kctd15a
cdh6
alx1
fam46ba
ved
crip2
nes
ovol1
sox1b
lhx5
ednrab
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

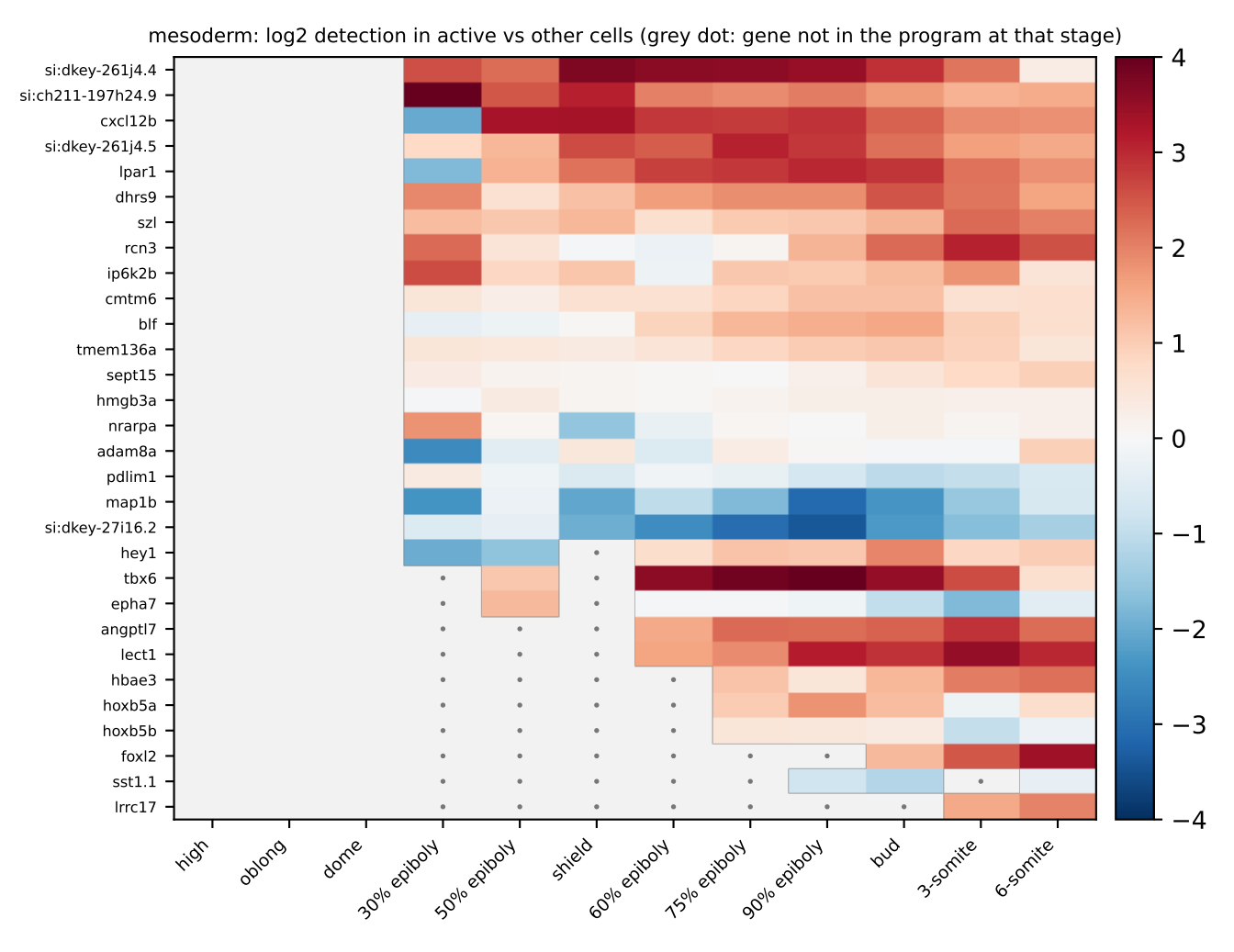

placodes: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
foxi1
ptgs2a
dlx4b
cldna
six4b
pitx3
tfap2c
sdpra
eya1
irf6
insm1b
pax8
pnocb
her13
foxj1b
fgf24
pih1d3
chodl
sp5a
tlx3b
scrt2
drgx
insm1a
sfrp2
atoh1b
isl2b
hmx3a
atoh1a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

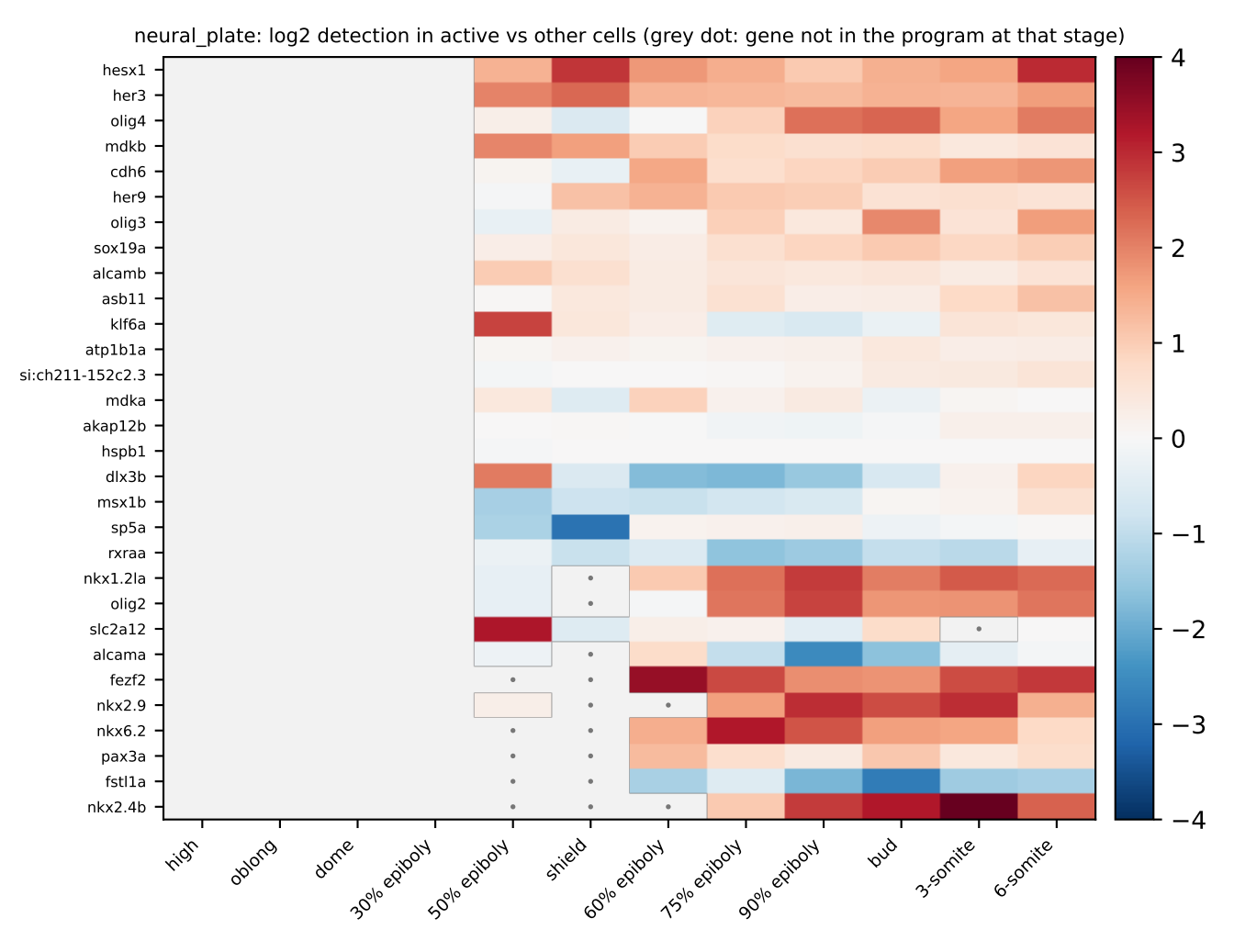

non_neural_ectoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
tp63
gata3
foxi3a
fermt1
tfap2c
bambib
grhl1
foxi3b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

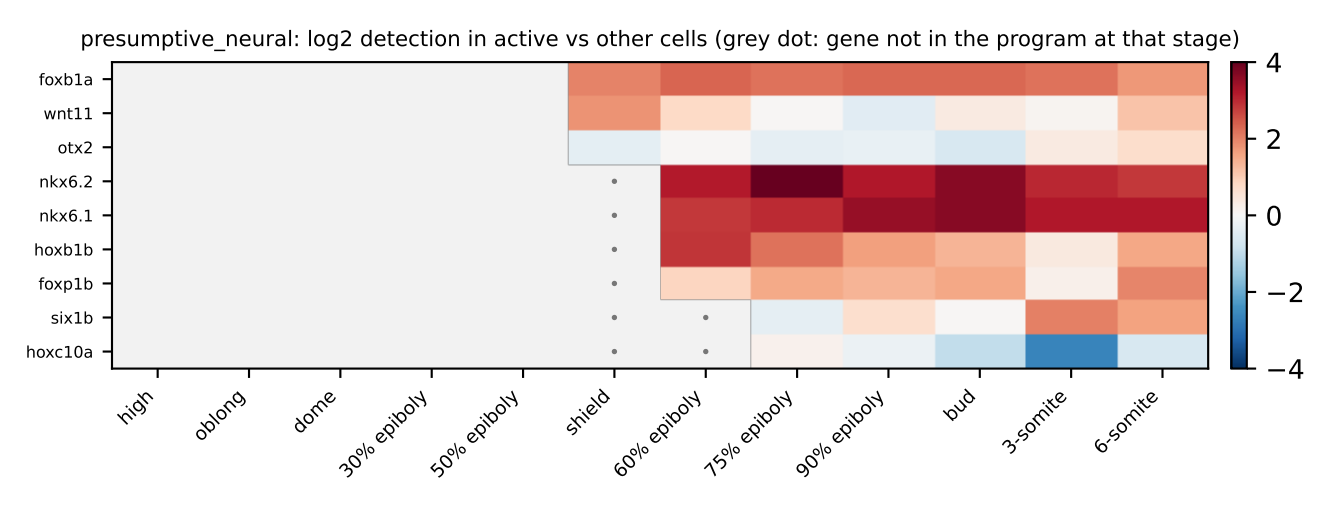

axial_mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
ntd5
ip6k2b
rcn3
ckap4
fam212aa
fam3c
pak1
perp
dnajb11
si:dkey-112a7.4
trib3
cnn2
acvr1ba
spty2d1
egln3
arg2
twist2
fbln2
creb3l1
tph1b
itga6a
bcam
slit2
hfe2
apobec2a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

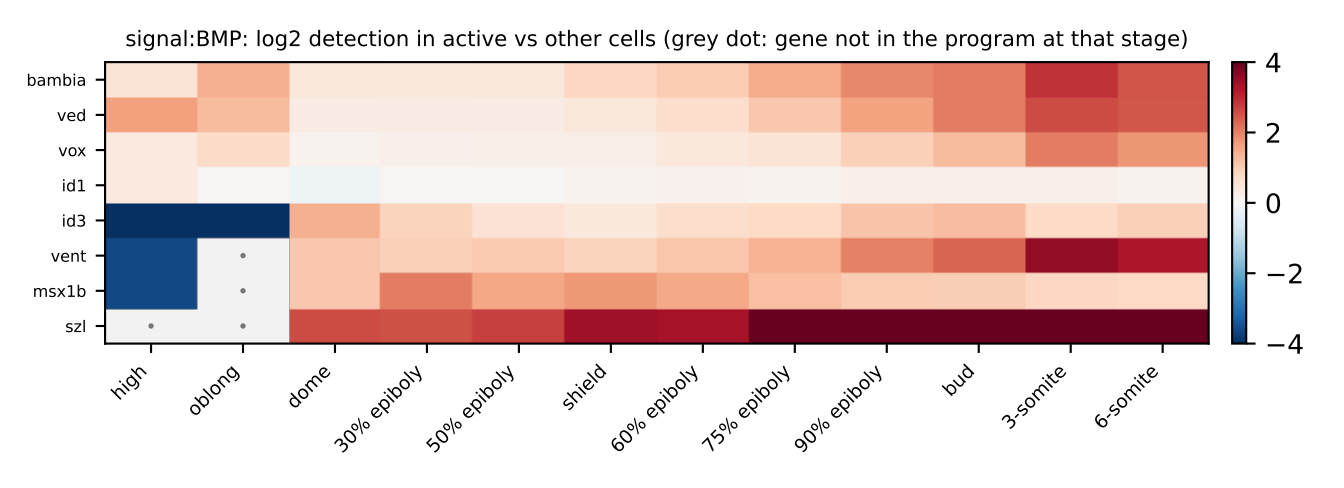

hatching_gland: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
zgc:162964
bhlha15
dnajc3a
frzb
cnpy1
rrbp1b
pitx2
cldnh
tagln2
he1a
si:ch211-153b23.5
stk10
hsp90b1
capns1a
inhbaa
fbxo2
bambia
gadd45ga
pfn1
hspa5
calr
pdia3
id3
calr3b
xbp1
hyou1
cd63
ssr4
sec61b
sec61g
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

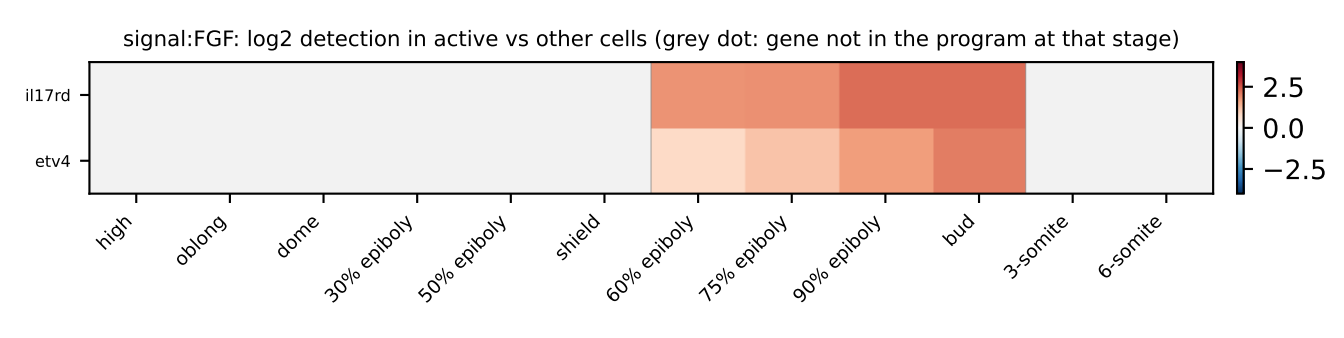

margin: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
mespab
lft1
cyp27c1
aplnrb
mixl1
gata6
wnt8a
lft2
p4ha2
flrt3
foxa
rdh10a
efnb2a
cdx4
eve1
arl4ab
rnd1l
hes6
dhrs3b
fscn1a
ved
sp5l
vox
xbp1
cdkn1ca
gadd45bb
mespaa
msgn1
serpinh1b
cpn1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

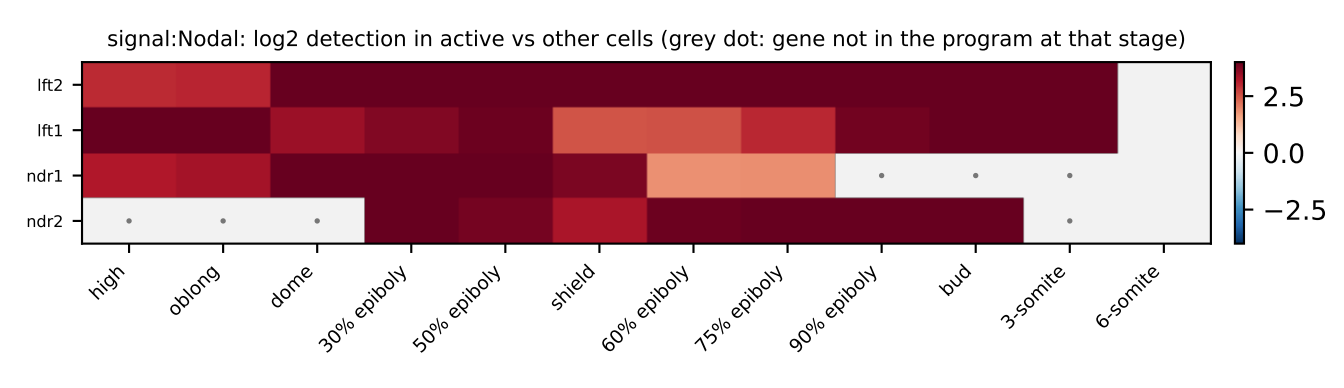

mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
si:dkey-261j4.4
si:ch211-197h24.9
cxcl12b
si:dkey-261j4.5
lpar1
dhrs9
szl
rcn3
ip6k2b
cmtm6
blf
tmem136a
sept15
hmgb3a
nrarpa
adam8a
pdlim1
map1b
si:dkey-27i16.2
hey1
tbx6
epha7
angptl7
lect1
hbae3
hoxb5a
hoxb5b
foxl2
sst1.1
lrrc17
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

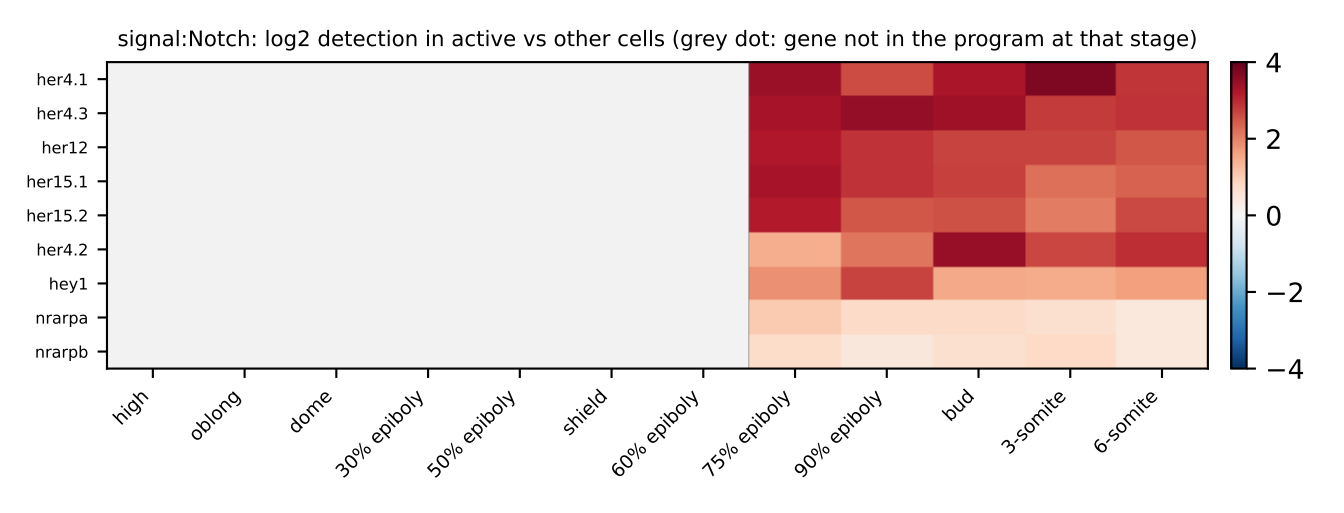

neural_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
hesx1
her3
olig4
mdkb
cdh6
her9
olig3
sox19a
alcamb
asb11
klf6a
atp1b1a
si:ch211-152c2.3
mdka
akap12b
hspb1
dlx3b
msx1b
sp5a
rxraa
nkx1.2la
olig2
slc2a12
alcama
fezf2
nkx2.9
nkx6.2
pax3a
fstl1a
nkx2.4b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

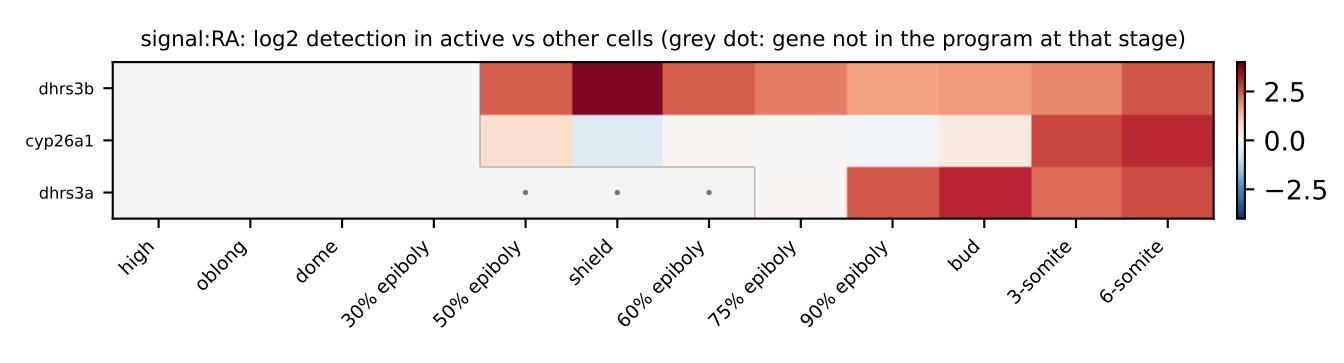

presumptive_neural: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
foxb1a
wnt11
otx2
nkx6.2
nkx6.1
hoxb1b
foxp1b
six1b
hoxc10a
4
2
0
−2
−4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

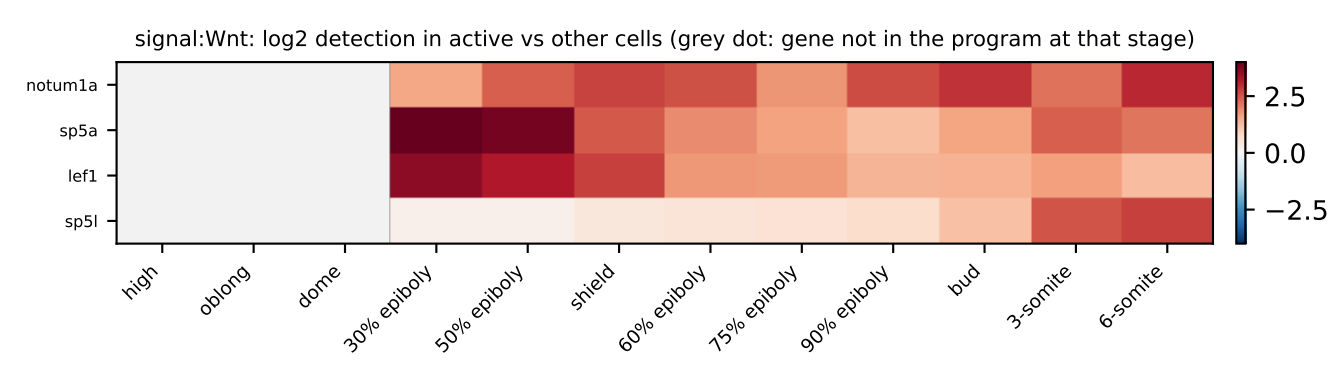

signal:BMP: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
bambia
ved
vox
id1
id3
vent
msx1b
szl
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

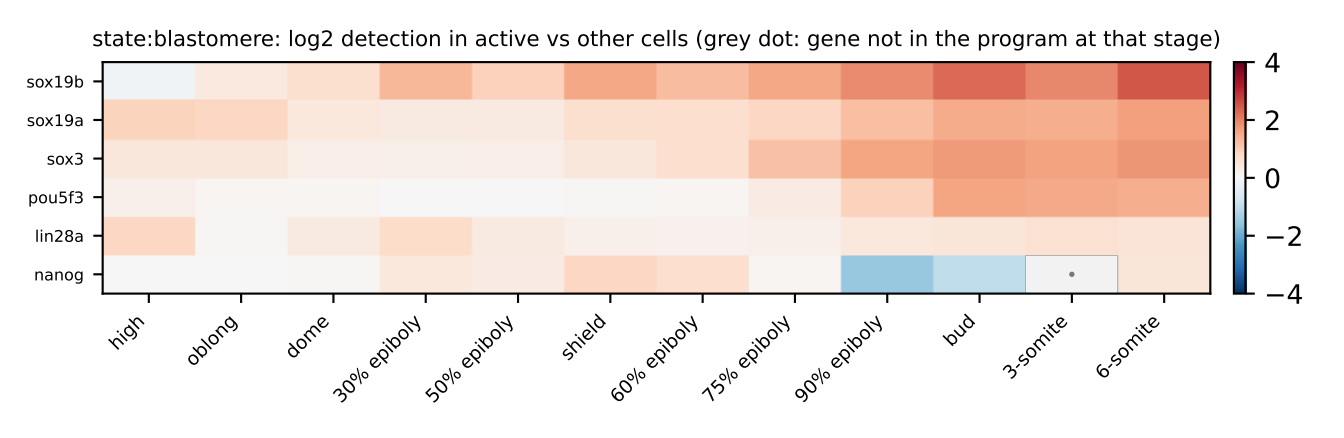

signal:FGF: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
il17rd
etv4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

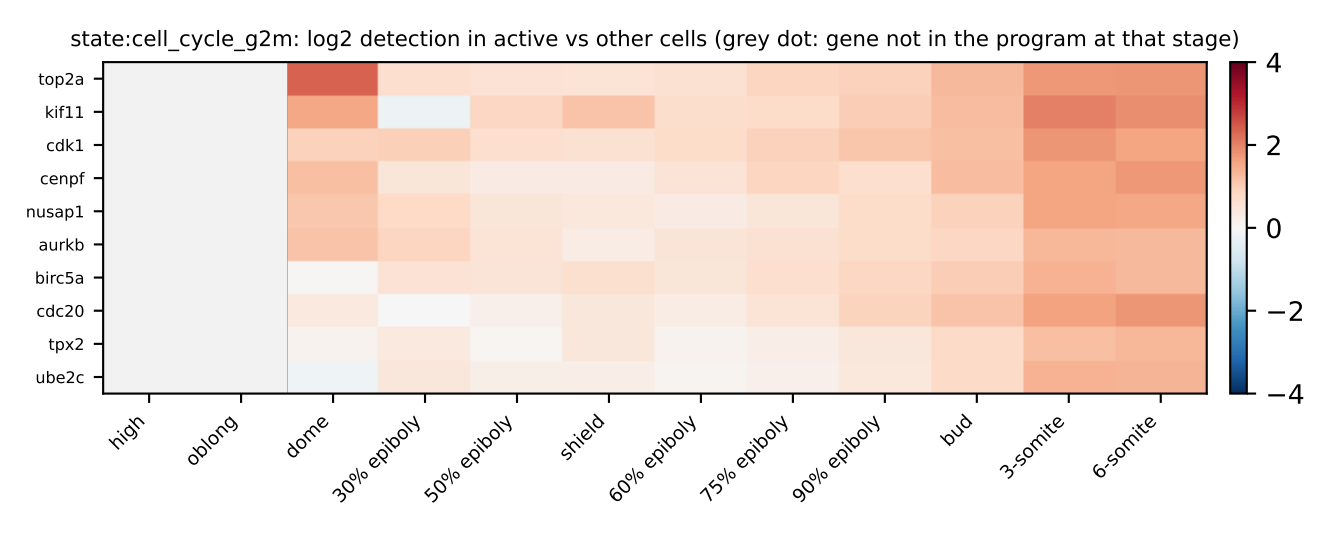

signal:Nodal: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
lft2
lft1
ndr1
ndr2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

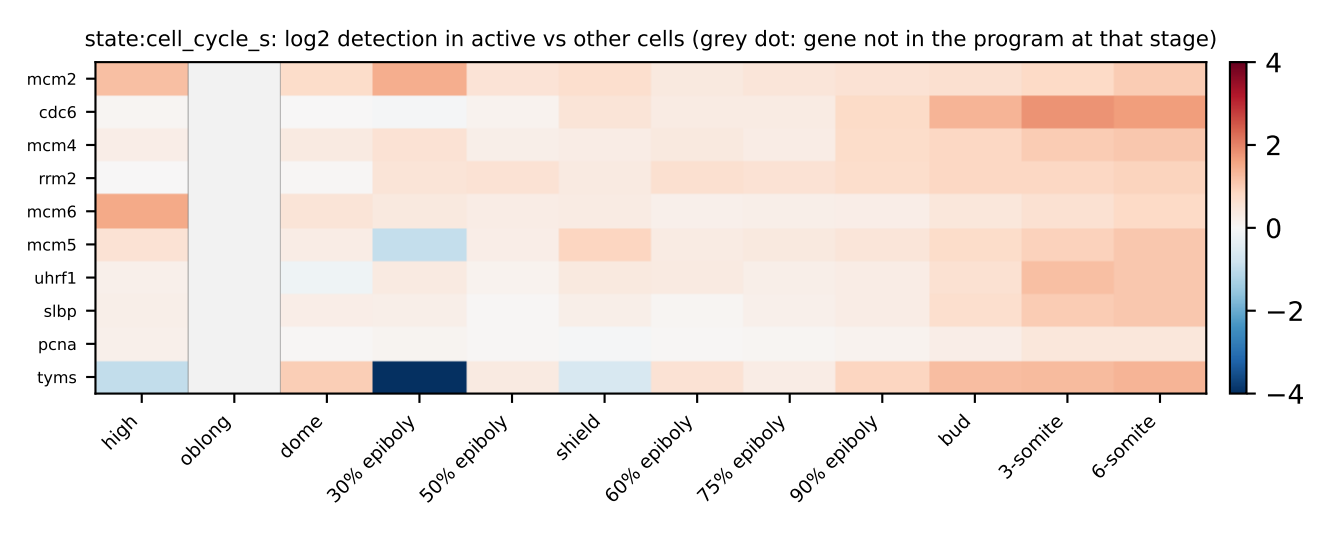

signal:Notch: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
her4.1
her4.3
her12
her15.1
her15.2
her4.2
hey1
nrarpa
nrarpb
4
2
0
−2
−4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

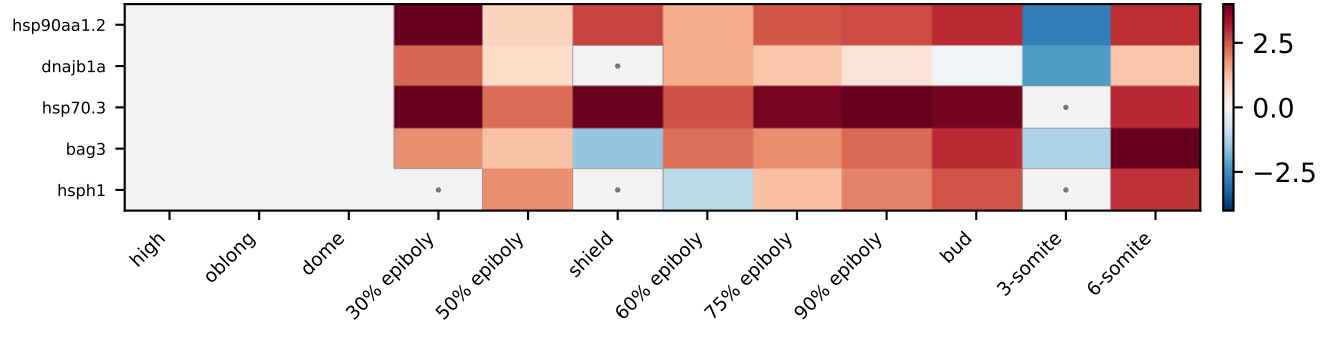

signal:RA: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
dhrs3b
cyp26a1
dhrs3a
2.5
0.0
−2.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

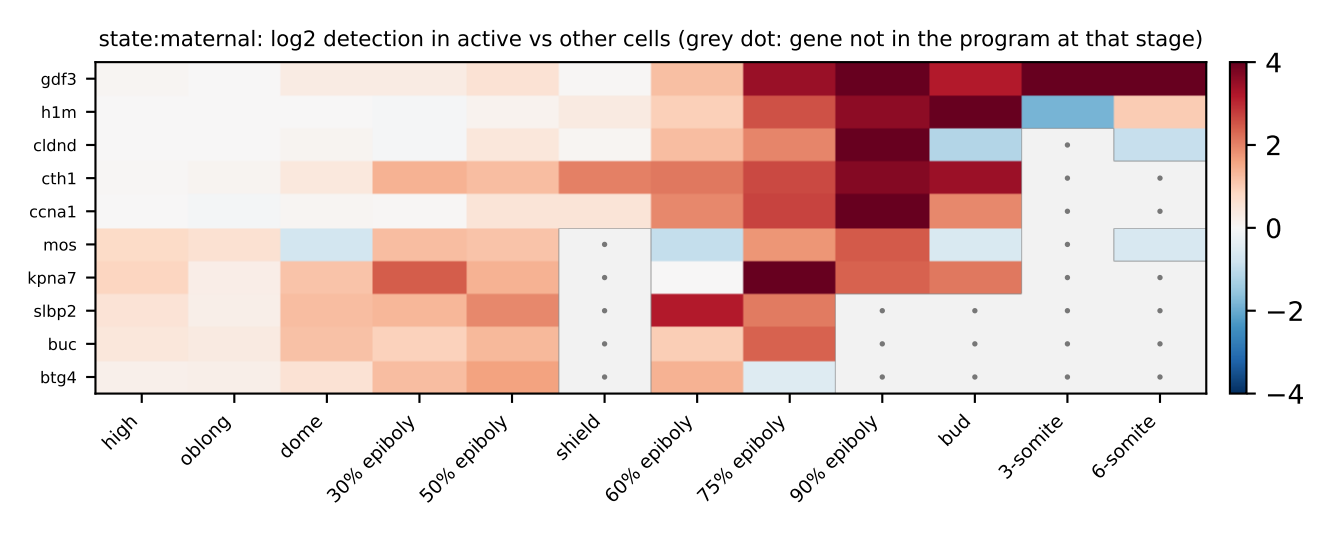

signal:Wnt: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
notum1a
sp5a
lef1
sp5l
2.5
0.0
−2.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

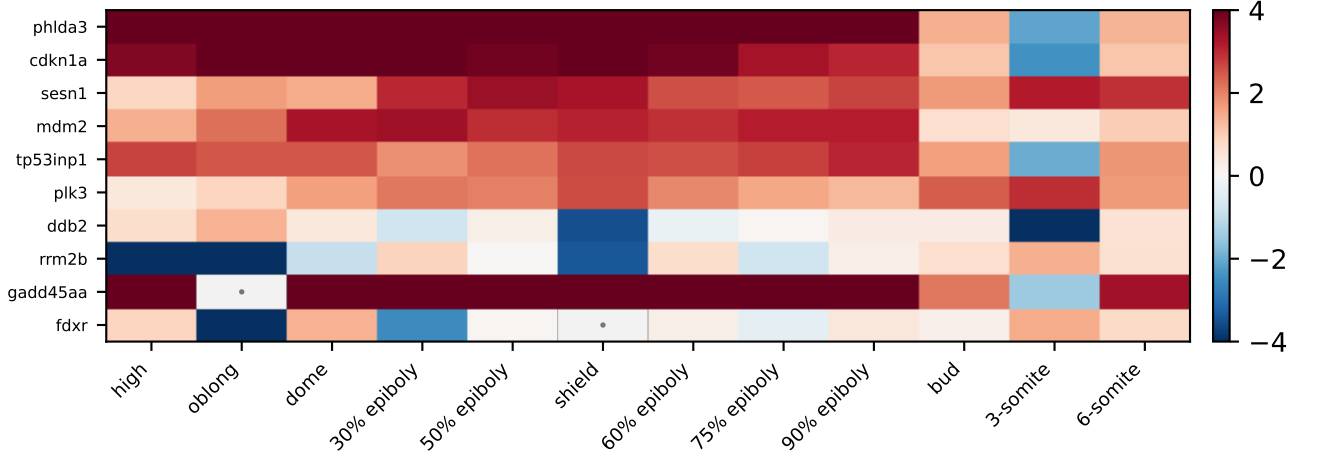

state:blastomere: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
sox19b
sox19a
sox3
pou5f3
lin28a
nanog
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:cell_cycle_g2m: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
top2a
kif11
cdk1
cenpf
nusap1
aurkb
birc5a
cdc20
tpx2
ube2c
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:cell_cycle_s: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
mcm2
cdc6
mcm4
rrm2
mcm6
mcm5
uhrf1
slbp
pcna
tyms
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:heat_shock: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
hsp90aa1.2
dnajb1a
hsp70.3
bag3
hsph1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5


state:maternal: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gdf3
h1m
cldnd
cth1
ccna1
mos
kpna7
slbp2
buc
btg4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:p53_response: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
phlda3
cdkn1a
sesn1
mdm2
tp53inp1
plk3
ddb2
rrm2b
gadd45aa
fdxr
4
2
0
−2
−4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite